

이학박사학위논문

Patterning by ion-beam sputtering
이온 다발 때려내기를 이용한 패턴 만들기

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조 민 응

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Abstract

We study the pattern formation on sputter-eroded surface both experimentally and theoretically. As an experimental approach, we perform ion beam sputtering on Au(001) at a grazing incidence of ion beam. First, single ion beam sputtering generates a ripple structure, and its morphology depends on several sputter conditions such as ion beam flux, fluence, energy, and azimuthal angle between the high symmetry axis of the substrate and ion beam projection. By controlling such sputter parameters, we can obtain various forms of ripple structure. A well-defined ripple structure requires sufficiently large flux to dominate degradation due to surface diffusion as well as match of crystal axis to ion-beam projection. Rather rich phases on this crystalline surface are obtained by a competition between ion-erosion kinetics and surface diffusion kinetics.

Second, we perform dual ion-beam sputtering (DIBS) to obtain more diverse patterns than those obtained from single ion-beam sputtering. Highly ordered two-dimensional nanopatterns are formed by DIBS: nanoholes in the diffusive regime and nanodots in erosive regime. Many salient features such as the slope asymmetry and an instability triggered by flux in the erosive regime can be obtained.

Third, we sputter a pre-ripples surface by crossing-ion-beam sputtering (CIBS). Ripple deformation in CIBS could provide valuable information on kinetics, which is involved in the sputtering. Contrary to the thermal smoothing in the Bradley-Harper case, the smoothing could be occurred via non-thermal processes such as redeposition and slope-dependent erosion. We observe the coarsening of initial rip-

ple with quite large coarsening exponent. The behavior is not explained by thermal coarsening. Thus, a ripple-mediated transport mechanism is suggested for an efficient mass transport mechanism. Also ripple coarsening in the erosive regime shows the crossover from a power law at the early stage to logarithmic scaling at the later stage. We suggest that the logarithmic behavior might be attributed to the redeposition due to roughened surface as sputtering prolongs.

For a theoretical approach, we set up a modified Kuramoto-Sivashinsky (KS) equation, responsible for the DIBS. The derived equation is inherently the same as the KS equation, although the coefficients are altered to reflect DIBS. It predicts a stable surface, but fails to describe the pattern or surface instability invoked by the DIBS.

To check the hypothesis of redeposition, we perform a large-scale molecular dynamics (MD) simulation, which mimics the CIBS. Even for a short simulation time, redeposition of sputtered atoms on the trough and shadowed slopes had a significant role in the evolution of surface morphology.

In summary, we have conducted novel sputter experiments using the dual ion beams on single crystal surface, and obtained rich surface morphology. We tried to find relevant description for the obtained experiment results theoretically. The pattern formation is beyond the scope of conventional approaches, and still remains to be explored theoretically.

Keywords: patterning, pattern formation, nanopatterning, sputtering, ion beam sputtering, nanostructure, Kuramoto-Sivashinsky equation, ripple, Bradley-Harper theory, Linear instability analysis

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Chapter 1

Introduction

Since the discovery of micro-ripples on a glass by ion bombardment (a breakthrough in changing the conventional way of thinking of sputtering, a destructive process) [1], ion-induced patterning has been rising steadily as an active research area. Understanding of its basic mechanism ‘*self-organized growth*’¹, responsible for the pattern formation under far-from-equilibrium conditions, has become matured throughout the explorations theoretically, ranging from molecular dynamics in atomic level to kinetic equations in macroscopic level as well as vast experiments done on various samples and under various conditions. Moreover, nanostructures provided by the ion irradiation techniques have wide and useful applications [2] so that this topic has received great interest from the scientific community.

The ultimate goal of this patterning is to make a perfect and uniform pattern on a submicron scale in a controllable manner, which itself is very demanding in our information age. This patterning achieves compact and highly-integrated optoelectronic devices such as optical storage media, magnetic recording media, LCD

¹“Any growth scenario is governed by the competition between kinetics and thermodynamics. We thus use the term ‘self-organized growth’ to describe autonomous order phenomena mediated by mesoscale force fields or kinetic limitations in growth processes, whereas ‘self-assembly’ is reserved for the spontaneous association of a supramolecular architecture from its molecular constituents. The term ‘self-organization’, in contrast to self-organized growth and self-assembly, is usually used in a different context, as it relates to dissipative structure formation in systems far from thermodynamic equilibrium and the initial emergence of biological macromolecules.” from Ref. [3]

display panels, and semiconductor devices.

The ion beam sputtering offers a viable tool for nanopatterning because it has many advantages such as low cost, high throughput (fast due to employing the self-organized way), and wide applicability regardless of materials. For instance, hexagonally well-ordered arrays of nanodots are formed remarkably uniform on the GaSb surface, the well-known semiconductor material, within just a single 1-minute dose by the ion beam sputtering at normal incidence [4]. Meanwhile, at off-normal angles of incidence, it has been reported that a periodic ripple structure is formed on various substrates ranging through insulators [5–7], semiconductors [8–12, 56], and metals [14–18]. However, there are still major limitations encountered as there are a number of defects and disorders of the pattern, which should be overcome to reach the level sufficient to a real application. To date, although many experimental and theoretical results for pattern formation on the sputter-eroded surface have been accumulated, the description of it is not fully satisfactory.

This dissertation is organized as follows: In Chapter 1, we briefly review relevant sputtering theories or models. That includes corresponding kinetic equations as well as scaling theory for generated surface. In Chapter 2, pattern formation on Au(001) under various sputtering conditions is studied. We show the order, shape and size dependence of the surface morphology as functions of ion fluence, energy, flux, and the azimuthal angle to target surface. A novel approach, dual ion beam sputtering (DIBS) and its result are introduced in Chapter 3. A square-shaped nanodot/hole is obtained by the DIBS. A flux-sensitive transition from dot to ripple has also occurred. In order to find a relevant description of the pattern evolution during the DIBS, we develop an extended version of the Kuramoto-Sivashinsky equation. Some nonlinear features in surface morphology are still under inspection, as to whether it is caused by nontrivial sputtering effects such as redeposition. These effects might well be unveiled at sputtering on a nonplanar surface initially. Thus in Chapter 4, we study the deformation of the initially rippled surface by ion beam

sputtering. We summarize our findings and come to the conclusion in Chapter 5.

1.1 Sigmund theory

Sigmund set up a set of kinetic equations describing energy transfer during ion sputtering process in which complex collision cascades of atoms are accompanied inside the bulk. However, the main result of the Sigmund theory is that it provides the energy distribution in an analytic function with which one can calculate the deposited energy at a point on the target surface. The ion sputtering process are summed up as follows: an incident ion penetrates into the bulk, stops in a certain range, and then dissipates its kinetic energy all around the region. Subsequently, some atoms at/near the surface can attain sufficient energy to break their bonding and be sputtered away. The Sigmund theory approximates the spatial distribution of the energy dissipation to the Gaussian function

$$\varepsilon(\mathbf{r}') = \frac{\epsilon}{(2\pi)^{3/2}\sigma\mu^2} \exp \left\{ -\frac{Z'^2}{2\sigma^2} - \frac{X'^2 + Y'^2}{2\mu^2} \right\}, \quad (1.1)$$

where Z' is the projected distance along the ion trajectory from a point P (where the ion stops) to a point O (where the deposited energy is calculated). X' and Y' are measured in the plane orthogonal to $\hat{Z}'$ (Fig. 1.1). The Gaussian distribution is characterized by the following widths: the σ along the incident-ion direction and the μ in its transverse direction. ϵ is the total energy carried by the ion [19].

While many ions arrive here and there on the target surface with flux J , only a few of them within the range $\mathcal{R}$ around a point O on the surface give the influence on the normal erosion velocity V_O , which is proportional to the total deposited energy at O given by

$$V_O = p \int_{\mathcal{R}} d\mathbf{r} \varepsilon(\mathbf{r}) f(\mathbf{r}), \quad (1.2)$$

where $\varepsilon(\mathbf{r})$ is the locally deposited energy on a point positioned $\mathbf{r}$ at the surface. $f(\mathbf{r})$ is the local ion flux, corrected from average ion flux J due to local surface curvature [20]. Note that $d\mathbf{r}$ is not a volume element $dXdYdZ$, but the surface element given by

$$d\mathbf{r} \simeq dXdY.$$

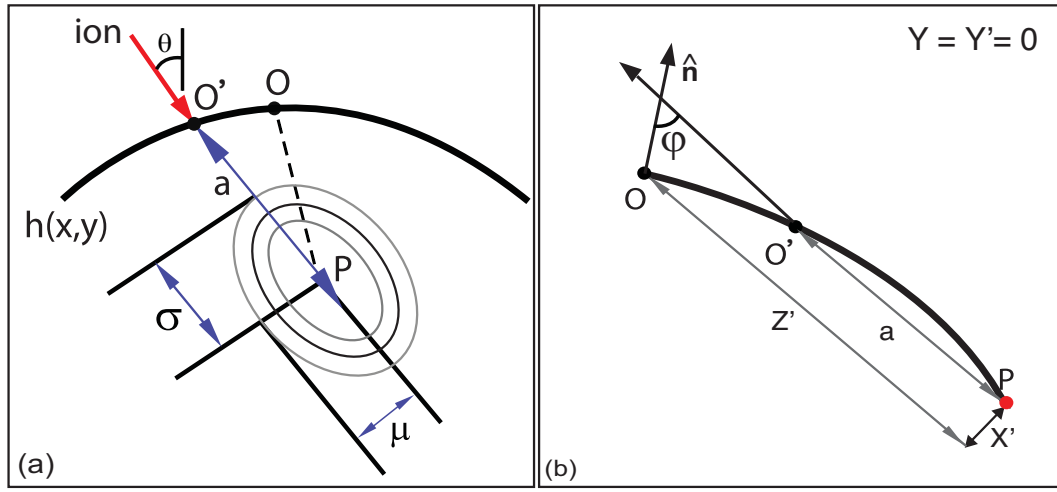


Figure 1.1. Schematic illustration for Sigmund theory. (a) His view replace complex ion-material interaction or collision cascade with energy dissipation from ion stopping following Gaussian function. Characteristic lengths of incident ions i.e. a , σ and μ represent penetration depth of ion and its longitudinal and transverse straggling, respectively. The concentric ellipsoids indicate contours of uniform energy. (b) X - Y coordinate is the local reference frame where the $\hat{Z}$ axis is parallel to the local normal to the surface $\hat{n}$. The $\hat{X}$ axis lies in the plane made by $\hat{Z}$ and $\hat{n}$ and then the $\hat{Y}$ axis indicates normal to this plane. Primed coordinate is a rotated frame with respect to the $Y = Y'$ direction in order to align the Z' axis parallel with the ion trajectory.

p is a material constant depending on the surface binding energy and scattering cross section given by

$$p = \frac{3}{4\pi^2} \frac{1}{NU_oC_o}, \quad (1.3)$$

where N is the target atom density [$\text{\AA}^{-3}$], U_o is the surface binding energy [$\text{eV}/\text{\AA}^3$], and C_o is a constant proportional to r_{eff}^2 , where r_{eff} is the effective radius of the interatomic potential [19].

1.2 Bradley-Harper theory

Based on the Sigmund theory, a continuum model has been proposed by Bradley and Harper (BH) [21], in which the height $h(x, y, t)$ of the eroded surface is described by a linear equation,

$$\partial_t h = \nu_x \partial_x^2 h + \nu_y \partial_y^2 h - K \nabla^4 h, \quad (1.4)$$

where ν_x and ν_y are the effective surface tensions in respective directions generated by the erosion process. Negative surface tension ($\nu < 0$) conducts the surface more rougher as the sputtering goes on. While normal surface tension ($\nu > 0$) leads the surface to smooth one. The BH theory took thermally-induced surface diffusion into account, so-called the Mullins' term [22, 23] in the height evolution equation as the eligible relaxation mechanism for ripple formation, independent of the Sigmund's theory. Without this additional lateral relaxation process, surface become rapidly rough and unstructured [24] as the sputtering proceeds. K is the relaxation rate representing isotropic smoothing of the surface due to thermal self-diffusion (diffusion of the atoms of which the surface is comprised), given by

$$K = \frac{D_s \gamma n}{N^2 k_B T}, \quad (1.5)$$

where γ is the surface free energy per area, i.e., the surface tension, n is the areal density of diffusing atoms, N is the atomic density of the substrate, and D_s is the

surface diffusion constant (surface diffusivity), which obeys an Arrhenius temperature dependence [21]

$$D_s = D_0 \exp\left(\frac{-\Delta E}{k_B T}\right). \quad (1.6)$$

Consequently, K has following T dependence (if n or γ is independent of (or negligibly dependent on) T):

$$K \propto \frac{1}{T} \exp\left(\frac{-\Delta E}{k_B T}\right), \quad (1.7)$$

indicating that it increases in an approximately exponential way as temperature increases.

Linear stability analysis

The Bradley-Harper equation, Eq. (1.4) is linear and translational invariant, thus can be solved by employing the Fourier transformation [25]. For simplicity, let us consider one dimensional version of the BH equation. Then the Eq. (1.4) becomes

$$\partial_t h = \nu \partial_x^2 h - K \partial_x^4 h, \quad (1.8)$$

where constant height h_0 is a definite solution. We introduce small perturbation $\zeta(x, t)$ into h i.e. $h(x, t) = h_0 + \zeta(x, t)$ and then expand the Eq. (1.8) to linear order in $\zeta(x, t)$. Now the equation is expressed by $\zeta(x, t)$

$$\partial_t \zeta = \nu \partial_x^2 \zeta - K \partial_x^4 \zeta. \quad (1.9)$$

This partial differential equation can be solved easily by the Fourier transformation. Introducing $\hat{\zeta}(k, t)$

$$\hat{\zeta}(k, t) = \int \frac{dx}{(2\pi)} \exp(ikx) \zeta(x, t), \quad (1.10)$$

the equation (1.9) is then transformed by

$$\begin{aligned} \frac{d\hat{\zeta}}{dt} &= -\nu k^2 \hat{\zeta} - K k^4 \hat{\zeta} \\ &= \omega(k) \hat{\zeta}, \end{aligned} \quad (1.11)$$

where $\omega(k) = -\nu k^2 - Kk^4$, and we assume that the spatial and temporal parts are separated by the ansatz $e^{ikx+\omega(k)t}$. The temporal solution of ζ is directly obtained from the Eq. (1.11), given by $e^{\omega(k)t}$. $\omega(k)$ has an role of the growth rate of the perturbation. If $\omega < 0$, the growth mode is stable because the perturbation exponentially decayed. Otherwise, the mode is unstable because the perturbation or amplitude grows exponentially. This explains the ripple formation and wavelength selection well: the second order derivative term contributes to growing the mode exponentially in small k (long distance in real space) but it is stabilized by fourth order derivative term. Thus the BH-instability competes with the smoothing mech-

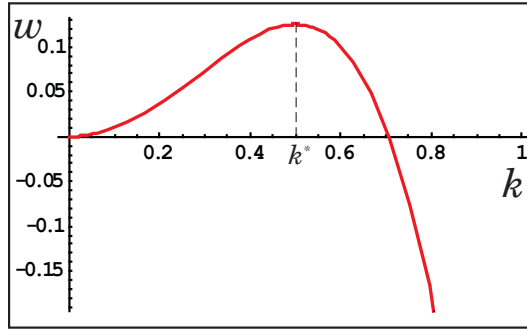


Figure 1.2. $\omega(k)$ versus k when $\nu = -1$ and $K = 2$ in Eq. (1.8).

anism. The characteristic wavelength determines from the maximum value of $\omega(k)$ in the unstable region, given by

$$k^* = \sqrt{\frac{\nu}{2K}}. \quad (1.12)$$

Because it is overwhelming than others in the sense of the exponential increase and survived in the early stage of morphology evolution. When height modulation is large enough, we encounter nonlinear mechanisms have crucial role in the pattern formation. According to the linear instability analysis, the balance of the erosion term $(-|\nu|\partial_i^2 h)$ and the surface diffusion term $(-K\nabla^4 h)$ generates ripples with

wavelength

$$\ell_i = 2\pi \sqrt{\frac{2K}{|\nu_i|}}, \quad (1.13)$$

where i refers to the direction x or y along which the associated ν_i is the largest. When the incidence angle θ is close to grazing (normal), the ripple wave vector becomes perpendicular (parallel) to the projection of the ion beam due to the angle and flux dependence of ν as shown in Fig. 1.3. According to Ref. [20,21], ν is given as a function of flux and incidence angle

$$\begin{aligned} \nu_x &= \frac{Fa}{2} \{2s^2 - c^2 - a_\sigma^2 s^2 c^2\}, \\ \nu_y &= -\frac{Fa}{2} c^2, \end{aligned} \quad (1.14)$$

with the definition (1.16, 1.17) referred in the following section.

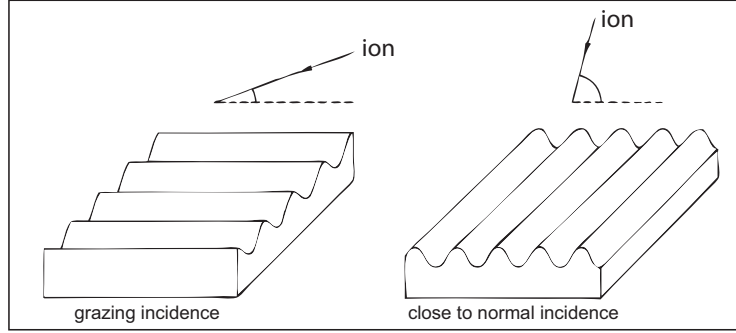


Figure 1.3. Ripple orientation depends on the incidence angle of ion. At a grazing incidence, the surface instability along the beam projection (x) disappears due to low flux. As a result, a ripple along y -axis is generated. At incidence close to normal, an instability in each direction is observed as similar as with the case at normal incidence. However, a ripple having wavevector parallel with x direction is observed when $\nu_x < \nu_y$ due to enough large a/σ [20].

Qualitative description of ripple orientation

Figure 1.4 shows a surface instability at off-normal incidence of ion. The energy deposited at O' in a trough or valley due to ion incidence at P' is larger than O

in a ridge or hilltop due to ion incidence at P' because the distance from center of corresponding elliptic curves is shorter in O' than in O . In each case, ion penetration depth is same. As a result, sputter yield is larger in a trough than in a ridge, which causes the surface roughness amplified; the surface gets rougher as ion sputtering continues. Surface instability arises! However, one should remember the fact that the deposited energy is also a function of local flux. At a grazing incidence, the surface instability along the beam projection (x) disappears due to low flux arrived. As a result, ripple orientation is changed according to the incidence angle of ion as shown in Fig. 1.3. Since an instability along y -direction remains (ν_y is always negative when ion beam is incident along x -axis [20].), a ripple is generated parallel with y -axis (wavevector perpendicular to x -axis). At incidence close to normal, an instability in each direction is observed as similar as with the case at normal incidence. However, a ripple having wavevector parallel with x direction is usually observed when $\nu_x < \nu_y$ due to enough large a/σ [20, 21]. The ripple orientation phase diagram is shown in Fig. 1.5.

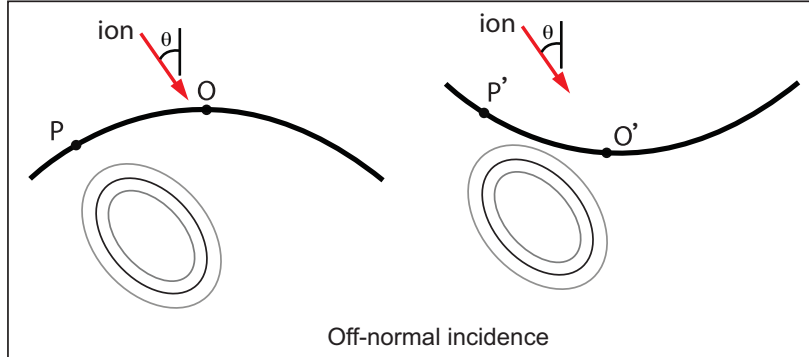


Figure 1.4. Ripple orientation at off-normal incidence of ion-beam. The energy deposited at O' due to ion incidence at P' is larger than O due to ion incidence at P' because the distance from center of corresponding elliptic curves is shorter in O' than in O . In each case, ion penetration depth is same. As a result, sputter yield is larger in a trough than in a ridge, and thus surface instability arises in the ion bombardment.

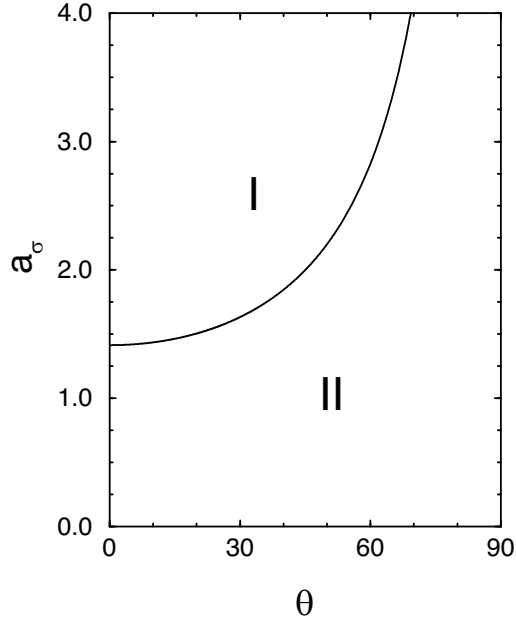


Figure 1.5. Ripple orientation phase diagram for the isotropic case $\sigma = \mu = 1$. $a_\sigma \equiv a/\sigma$. Region I: $\nu_x < \nu_y < 0$; Region II: $\nu_x > \nu_y$. This is calculated from Eq. 1.14. Ref. [20].

1.3 Kuramoto-Sivashinsky equation

Cuerno and Barabási derived a stochastic nonlinear equation to derive the evolution and scailing properties of surfaces eroded by ion bombardment [26]. Their work is an more complete extension of the BH theory. They expanded out the slope dependence of the sputter yield to second order in slope, and introduced noise as relevant physical process to mimic stochastic arrival of ions. While the linear BH theory is quite successful in predicting the ripple wavelength and its orientation [27], it cannot explain a number of experimental features such as

- the behavior of the wavelength of the ripple, that is independent of the temperature in the low-temperature regime
- the saturation of the ripple amplitude [29, 30, 71]
- the observation of the rotated ripple [14]

- the appearance of kinetic roughening [31, 32]

These phenomena can be explained by the noisy nonlinear equation called the anisotropic Kuramoto-Sivashinsky (KS) equation [20, 26]

$$\begin{aligned} \partial_t h = & \nu_x \partial_{xx}^2 h + \nu_y \partial_{yy}^2 h + \frac{\lambda_x}{2} (\partial_x h)^2 + \frac{\lambda_y}{2} (\partial_y h)^2 \\ & - D_{xx} \partial_x^4 h - D_{xy} \partial_x^2 \partial_y^2 h - D_{yy} \partial_y^4 h - K \nabla^4 h + \eta(x, y, t), \end{aligned} \quad (1.15)$$

where λ_x and λ_y describe the tilt-dependent erosion rates in each direction. D_{xx} , D_{yy} , and D_{xy} are the ion-induced effective diffusion constants. $\eta(x, y, t)$ is the uncorrelated white noise with zero mean, mimicking the randomness resulting from the stochastic nature of ion arrival at the surface [26, 33]. At low temperatures, the thermal diffusion term is negligible due to Eq. (1.7) while the ion-induced diffusion term $D_{ij} \partial_i^2 \partial_j^2 h$ ($i, j \in x, y$) is significant, and vice versa in the high temperature limit. Other coefficients in Eq. (1.15) (ν_i , D_{ij} , $\lambda_{i,j}$) depend on the experimental parameters such as the ion-beam energy ϵ , flux J , the incidence angle from surface normal θ , and the ion-material characteristics (a, σ, μ). In any case, (ν_i , D_{ij} , $\lambda_{i,j}$) are proportional to ϵ and J [20].

Nonlinear features of the KS equation

For the sputter-erosion case, it has $\nu < 0$ and $D > 0$, while the signs of λ_x and λ_y vary as the incidence angle of the ion beam. When the nonlinear terms and the noise term are neglected, Eq. (1.15) reduces to the linear one as Eq. (1.4) and predicts ripple formation. It is known that the anisotropic nonlinear KS equation ($\nu_x, \nu_y < 0$, $D_{xx} = D_{yy} = D_{xy}/2$, and $\lambda_x = \lambda_y$) allows ripple formation at an early stage, and then asymptotically (for large time and length scales) predicts kinetic roughening with exponents similar to those obtained by experiments (Fig. 1.6) [31, 34].

It is known that for positive ν_x and ν_y , Eq. (1.15) reduces to the anisotropic Kardar-Parisi-Zhang (KPZ) equation [55, 69], whose scaling behavior is controlled

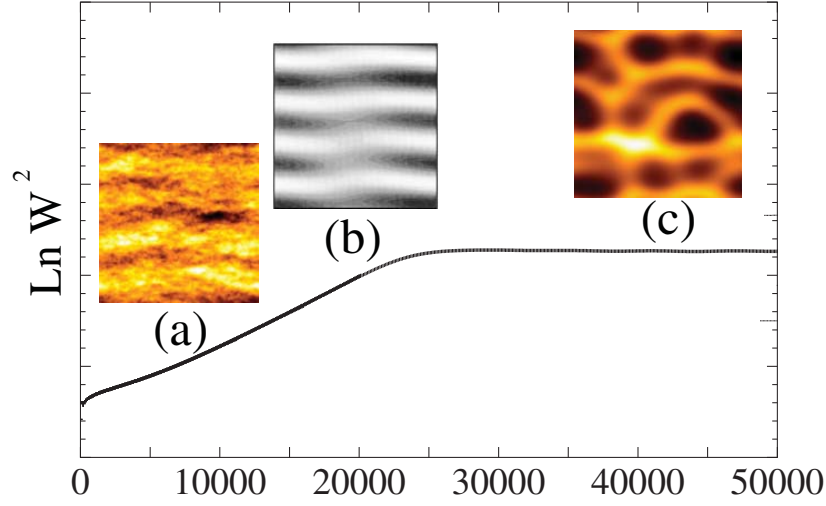


Figure 1.6. Plot of a surface with the parameter sets: (a) The surface configuration at time $t = 7 \times 10^3$, (b) at time $t = 2 \times 10^4$, and (c) at time $t = 5 \times 10^5$, which shows kinetic roughening. Following parameter set is used: $\nu_x = -0.0001$, $\nu_y = -0.6169$, $K_x = K_y = K_{xy}/2 = 2$, and $\lambda_x = \lambda_y = -0.001$. Eq. (1.13) predicts the wavelength $\ell_y \approx 16$. The roughness initially grows exponentially and saturates, and then kinetic roughening exhibits on the surface. Ref. [34].

by the sign of $\lambda_x \cdot \lambda_y$ [35]. For sputter-etched surface (at least one ν_i has negative value), the case when $\lambda_x \cdot \lambda_y < 0$ was investigated by Rost and Krug [36]. Their integration of the noiseless version of Eq. (1.15) provided an evidence that when $\lambda_x \cdot \lambda_y < 0$, a novel ripple structure, unaccounted for by the linear theory, emerges and its orientation is rotated with respect to the ion projection. The nonlinear effects have been largely unexplored experimentally due to the lack of theoretical predictions for experimentally detectable signatures that are clearly distinctive in comparison with those according to the linear theory.

The scaling behavior depends on the relative signs of ν_x , ν_y , λ_x and λ_y . The variations of these coefficients as functions of a_σ and θ lead to the phase diagram shown in Fig. 1.8 [26]. In the Region I and II, a ripple structure dominates the surface morphology as discussed by the BH theory. The large length scale (far larger

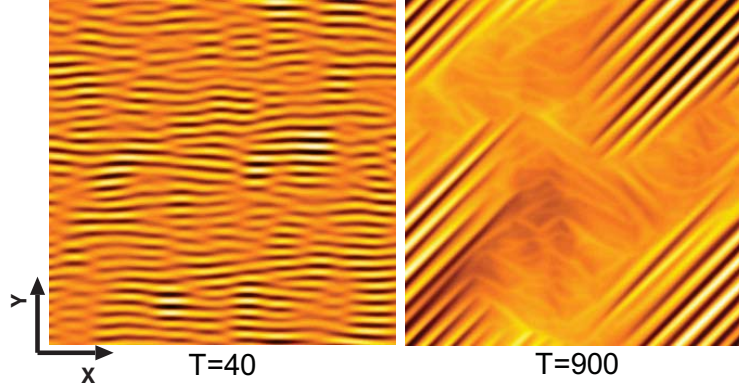


Figure 1.7. A rotated ripple (right) reveals at long time by the interplay with nonlinear terms. Here, we used following coefficients in the KS equation: $\nu_x = 0.01$, $\nu_y = -1$, $D_{xx} = 1$, $D_{yy} = 1$, $D_{xy} = 2$, $\lambda_x = 10^{-7}$, and $\lambda_y = -10^{-7}$.

than the wavelength) behavior is expected to be different because both nonlinearities and the noise may become relevant. The exact scaling properties of Eq. (1.15) at large scale are not known yet. However, one can consider an analogy with the anisotropic KPZ (AKPZ) equation [35]. Then one would observe algebraic scaling in region I, where both λ_x and λ_y has the same (negative) sign. While in region II ($\lambda_x \cdot \lambda_y < 0$), a logarithmic scaling would be expected if the AKPZ mechanism works. In the Region III, a periodic structure associated with the instability is directed along the y direction is expected due to $\nu_x > 0$ and $\nu_y < 0$. Again, little is known about the scaling behavior. If ν_y renormalizes to a positive value, and that the AKPZ mechanism operates, one would obtain logarithmic scaling due to $\lambda_x \cdot \lambda_y < 0$.

1.3.1 Dependence of surface morphology on physical parameters

Energy dependence of wavelength λ

At high temperatures, thermally activated surface diffusion is a main relaxation mechanism since K increases with T as $K \propto \frac{1}{T} \exp(\frac{-\Delta E}{k_B T})$, so the wavelength is

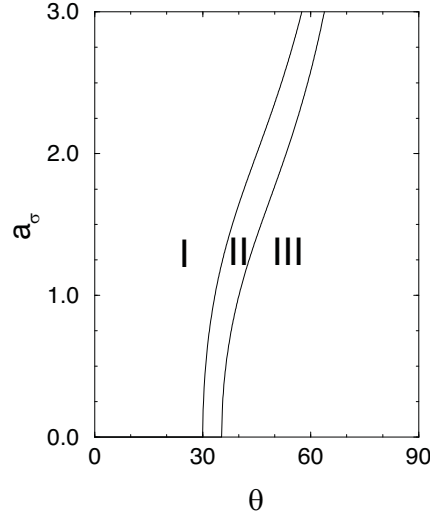


Figure 1.8. Phase diagram for the isotropic case $\sigma = \mu = 1$. Region I: $\nu_x < 0, \nu_y < 0, \lambda_x < 0, \lambda_y < 0$; Region II: $\nu_x < 0, \nu_y < 0, \lambda_x > 0, \lambda_y < 0$; Region III: $\nu_x > 0, \nu_y < 0, \lambda_x > 0, \lambda_y < 0$. Ref. [20, 26].

expressed by Eq. (1.13). The denominator $|\nu|$ is proportional to Fa where F is a global factor which can be extracted from all coefficients in the Eq. (1.15) except K that is given by

$$F \equiv \frac{J\epsilon pa}{\sigma\mu\sqrt{2\pi}f} e^{-a_\sigma^2 a_\mu^2 c^2 / 2f}, \quad (1.16)$$

with following numbers [20],

$$\begin{aligned} a_\sigma &\equiv \frac{a}{\sigma}, a_\mu \equiv \frac{a}{\mu}, \\ s &\equiv \sin \theta, c \equiv \cos \theta, \\ f &\equiv a_\sigma^2 s^2 + a_\mu^2 c^2. \end{aligned} \quad (1.17)$$

Also, F can be expressed by the local sputtering yield $Y(\theta)$ as

$$F = \frac{JaY(\theta)}{N}, \quad (1.18)$$

where N is atomic density of the target material. The ion penetration depth a has dependence on the incident ion energy ϵ : it would be proportional to the ion energy.

The more energy ion has, the deeper the ion would reach inside the bulk because of its large kinetic energy. Based on the Sigmund's theory of sputtering, following relation is given by

$$a(\epsilon) = \frac{1-m}{2m} \gamma^{m-1} \frac{\epsilon^{2m}}{NC_m}, \quad (1.19)$$

where C_m is a constant depending on the interatomic potential, and m is an exponent depending on the energy range,

$$m = m(\epsilon) \simeq \begin{cases} 1 & \text{for } \epsilon > 100 \text{ keV}, \\ 0.5 & \text{for } 10 < \epsilon < 100 \text{ keV}, \\ 0 & \text{for } \epsilon < 10 \text{ keV}. \end{cases} \quad (1.20)$$

Thus from $a \propto \epsilon^{2m}$ and $F \propto (\epsilon a)/(\sigma \mu) \sim \epsilon^{1-2m}$ (if $\sigma \sim \mu \sim a$), one can obtain following relation between wavelength and ion energy

$$\lambda \propto \epsilon^{-1/2}, \quad (1.21)$$

indicating that the ripple wavelength is decreased as the ion energy increases. We note that under the BH theory or the KS equation², the wavelength is independent of the sputter time t or the ion influence Ψ . Because the Eq. (1.4), linearly unstable, gives exponentially growth Fourier modes among which one particular Fourier mode k_c , the fastest growth mode, dominates over the others and emerges as the ripple wavelength.

Flux dependence of wavelength λ

From the Eq. (1.16), F is linearly proportional to the ion flux J . As a result, the wavelength, given by the Eq. (1.13), has same dependence with the ion energy,

$$\lambda \propto J^{-1/2}. \quad (1.22)$$

²In the initial stages of the morphological evolution, when the height modulation is small, we can simplify the evolution equation to the linear one. Thus even in the KS equation, the wavelength is selected by the linear part of the equation.

Therefore the wavelength decreases with increasing flux. Above we assume that the surface relaxation rate K is independent of both the projectile flux and the ion energy. However, one may notice that K could be influenced by sputtering process: the increase of ion-beam flux and energy can cause the increase of areal density of mobile adatom [59] or decrease of the effective temperature of the surface [63], which all involved in the surface diffusion through Eq. (1.5).

1.4 Damped Kuramoto-Sivashinsky equation

Recently, some improvements in the theoretical understanding of pattern formation on a sputter-eroded surface have been achieved. Facsko *et al.* [37] introduced the damped Kuramoto-Sivashinsky equation (DKS) for describing sputter-induced patterns, given by

$$\partial_t h = -v_0 - \alpha h + \nu \nabla^2 h - D \nabla^4 h + \frac{\lambda}{2} (\nabla h)^2 + \eta. \quad (1.23)$$

Here the $-\alpha h$ term, having exponentially decaying solution in time by itself, is combined to the noisy isotropic KS equation to take into account for redeposition or lateral redistribution of the sputtered atoms. This is a nontrivial microscopic mechanism beyond the Sigmund theory of the sputtering: among sputtered atoms, a larger amount of atoms could be captured at the surface when they come from valleys than crests, especially for the case of a highly corrugated surface. This kinetics are considered as the damping term with its simplest form, the linear order correction for the variation of $-v_0$.

Even though the microscopic origin of the damping term is not clearly derived, the DKS equation quite successfully reproduces experimental results, dot arrays on GaSb [4, 37] and InP [38] surfaces, even including their various types of defects, with the damping parameter in a narrow range close to $\alpha = 0.25$.

1.5 Hydrodynamic theory

A new type of continuum model has been introduced for surface patterning [39, 40], inspired from the so-called hydrodynamic theory of ripple formation in aeolian sand dunes [41–43, 86]. It consists of two coupled fields, height of the bombarded substrate $h(\mathbf{r}, t)$ and thickness of mobile surface adatoms layer $R(\mathbf{r}, t)$. They are coupled with each other as follows:

$$\frac{\partial R}{\partial t} = (1 - \phi)\Gamma_{\text{ex}} - \Gamma_{\text{ad}} - \nabla \cdot \mathbf{J}, \quad (1.24)$$

$$\frac{\partial h}{\partial t} = -\Gamma_{\text{ex}} + \Gamma_{\text{ad}}, \quad (1.25)$$

where Γ_{ex} (Γ_{ad}) is the rate of atom excavation (addition) from (to) the immobile bulk. The $\mathbf{J}$ term in Eq. (1.24) describes motions or currents of the mobile adatoms. Explicitly, $\mathbf{J} = -D\nabla R$ is given to take the isotropic thermal diffusion into account, where D is an isotropic diffusion constant. In analogy with the Gibbs-Thompson relation, Γ_{ad} is written by

$$\Gamma_{\text{ad}} = \gamma_0 [R - R_{\text{eq}}(1 - \gamma_{2x}\partial_x^2 h - \gamma_{2y}\partial_y^2 h)], \quad (1.26)$$

where γ_0 is the mean nucleation rate of a flat surface. γ_{2x} and γ_{2y} describe the dependence of the nucleation rate on the surface curvature. The rate of excavation Γ_{ex} is closely related with the experimental parameters such as incident angle, ion energy, flux, and other sputter conditions (e.g. rotating substrate [44]). Thus, it can be written in similar form with Eq. (1.15), given by

$$\begin{aligned} \Gamma_{\text{ex}} = \alpha_0 [& 1 + \alpha_{1x}\partial_x h + \alpha_{2x}\partial_x^2 h + \alpha_{2y}\partial_y^2 h \\ & + \alpha_{3x}(\partial_x h)^2 + \alpha_{3y}(\partial_y h)^2 \\ & - (\partial_x h)(\alpha_{4x}\partial_x^2 h + \alpha_{4y}\partial_y^2 h)]. \end{aligned} \quad (1.27)$$

The significance of this approach is based on the assumption that there is a finite fraction $\bar{\phi} \equiv 1 - \phi$ of redeposited atoms, and a nonzero average fraction of mobile

atoms R_{eq} even in the absence of redeposition during the sputtering process. These additional mechanisms can allow richer dynamics of the pattern growth beyond the limitations of the KS equation [39, 40].

1.6 A continuum equation for multiple ion sources

So far we focused on pattern formation induced by single ion-beam sputtering (where a unidirectional ion flux from source to target is used). To obtain more diverse and thus maybe functionally distinct nanostructures from typical dots and ripples, Carter [45] has proposed the use of dual ion-beam sputtering (DIBS), and furthermore multiple ion-beam sputtering. During the DIBS, two ion beams, which are perpendicularly crossing each other in the (x, y) plane, are simultaneously projected onto the substrate. It was suggested that the two individual ion flux may interfere to produce more advanced patterns with easily controllable physical parameters such as ion flux, ion energy, and incidence angle of the ion beam. Even though there has been no reported experimental work to embodying this idea, Vogel and Linz [46] numerically studied a sputter-eroded surface induced by four simultaneously incident ion beams, crossing each other perpendicularly in the (x, y) plane. Equipped with a generalized version of the DKS equation

$$\partial_t h = -(\alpha + \nabla^2)h - (\partial_x^4 + \delta \partial_x^2 \partial_y^2 + \partial_y^4)h + (\nabla h)^2 + \eta, \quad (1.28)$$

they obtained square patterns and superstructure arrays under $\delta > 2$ and $\alpha = 0.24$ (close to the instability $\alpha = 0.25$), served as a model for clearly representing the presence of ion-induced effective surface diffusion.

1.7 Scaling theory

Diverse interfaces in nature ranging from mountain terrain to nanoscale topography of solid films look just irregular, being far from their equilibrium shapes. However,

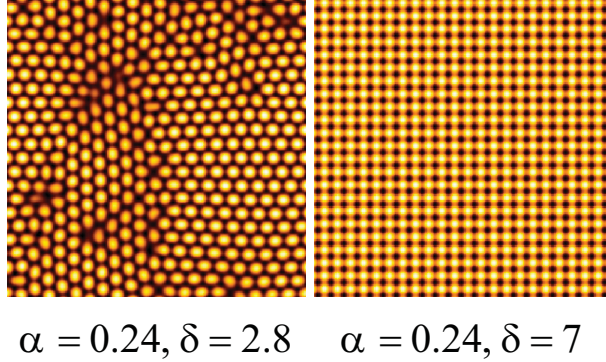


Figure 1.9. Morphology reproduced from Vogel and Linz's equation [46].

their roughness can be effectively described by the fractal scaling [47]. In far-from-equilibrium conditions, self-affine rough surfaces have been observed in many experiments such as film growth and sputtering even though their physical processes are significantly different [24, 48].³

To quantify the rough surfaces generated under many nonequilibrium processes, the scaling theory was suggested. One of quantities to characterize the surfaces is the surface roughness, the (vertical) fluctuation of surface height, which is defined by

$$W(L, t) \equiv \sqrt{\frac{1}{L^2} \sum_{\mathbf{r}} [h(\mathbf{r}, t) - \bar{h}]^2}, \quad (1.29)$$

where $\bar{h}$ is the average height given by

$$\bar{h} = \frac{1}{L^2} \sum_{\mathbf{r}} h(\mathbf{r}, t). \quad (1.30)$$

The root-mean-square (rms) width W is not sufficient to define the surface structure uniquely. Information in lateral direction is required for a complete description

³In film growth under equilibrium condition there are well-known growth modes such as *Frank-van der Merwe* (where the film grows layer by layer), *Stranski-Krastonov* (where the film initially grows in a layer-by-layer fashion, then reaches certain thickness and starts to build three dimensional structures) or *Volmer-Weber* (where three dimensional mound structure develops from its beginning, without layers forming in advance), and induced surface morphology cannot be described by the scaling concepts.

of the surface. We can usually choose it by the lateral correlation length ξ , within which the surface height of any two points are correlated. In the beginning, deposition or erosion events are uncorrelated, occurred locally at random sites. However, as the processes proceed, the event at one site is no longer isolated, but can interact with the neighbor events although the dynamics is still local. Thus ξ grows with time before crossover or saturation time t_c . Since in a finite system, ξ cannot exceed system size L , if it reaches the size of system ($\xi \sim L$), entire events are correlated, resulting in saturated W . The rms interface width W thus depends on system size L and scales as

$$W(L) \propto L^\alpha, \quad (1.31)$$

where the exponent α ($0 < \alpha < 1$) is called the ‘*roughness exponent*’, which illustrates short range properties of the surface. The smaller (closer to 0) it has, the more jagged or wiggled local surface emerges. This is because α characterizes how W depends (scales) on (as) L . In short-range length scale, for instance, if L is less than 1 unit distance, surface roughness increases (looks more largely fluctuating) as the roughness exponent increases. In order to compare the time evolution of the roughness, one should fix the system size for all cases and then obtain each roughness.

Time dependence of W is represented by the ‘*growth exponent*’ β , according to

$$W(L, t) \sim t^\beta. \quad (1.32)$$

In 1985, Family and Vicsek introduced the *dynamic scaling* concept in order to combine both spatial and temporal scaling behaviors [52]. Within this context, above relations on W are converged in the following scaling relation (Fig. 1.10),

$$W(L, t) \equiv t^\beta g\left(\frac{t}{L^z}\right), \quad (1.33)$$

where g is called a scaling function that satisfies following conditions,

$$g(u) \simeq \begin{cases} 1 & \text{for } u \ll 1, \\ u^{-\beta} & \text{for } u \gg 1. \end{cases} \quad (1.34)$$

The equation (1.34) implies that the scaling exponents α , β , and z are related to each other: when $t > t_c$ (corresponding to $u \gg 1$ above), $W = L^{z\beta} = t_c^\beta$. This gives $z = \alpha/\beta$, referred to as the ‘*dynamic exponent*’ and $\xi \propto t^{1/z}$ implicitly.

Numerical values of the exponents α , β , and z for various models introduced in this and following chapters, are summarized in table 1.1.

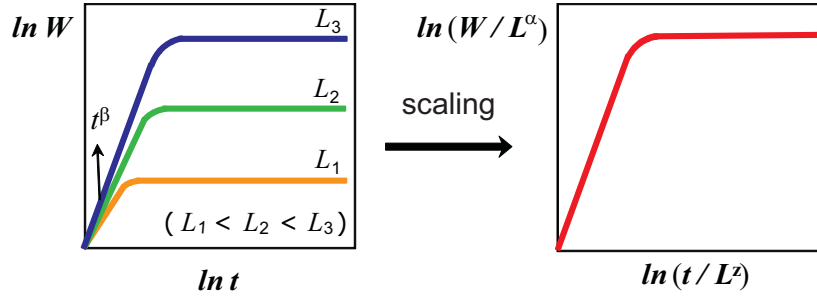


Figure 1.10. The left figure shows interface width for three different system size L increases according to the power law t^β and then saturates at L^α . Through the scaling relation (1.34), these three curves collapse into a single curve, called the scaling function g .

Table 1.1. Summary of the scaling exponents, predicted by various models for spatial dimensions $d = 2$. The DKS and hydrodynamic model has not yet fully studied their scaling behaviors. However, power law and logarithmic coarsening was reported in the hydrodynamic model [39].

Model	α	β	$1/z$	Ref.
Random deposition (RD)	–	0.5	–	[24]
Zeno model (1-D model for infinite step-edge barrier)	–	0.5	–	[53]
Edwards-Wilkinson (EW)	$(2-d)/2 = 0$	$(2-d)/4 = 0$	0.5	[25]
Mullins (surface diffusion corresponding to $\nabla^4 h$)	$(4-d)/2 = 1$	$(4-d)/8 = 0.25$	0.25	[24]
Kardar-Parisi-Zhang (KPZ)	0.38	0.024	0.63	[24]
KS (early time)	$0.75 - 0.8$	$0.22 - 0.25$	$0.25 - 0.33$	[79]
KS (late time)	$0.25 - 0.28$	$0.16 - 0.21$	–	[79]

Chapter 2

Pattern formation by single ion-beam sputtering

At normal incidence of ion-beam, the ion-induced pattern on target surface should have isotropic symmetry; It should not show any preferential direction if the target surface is amorphous. However, the symmetry could be broken when the incidence angle of ion-beam is deviated from surface normal; A periodic structure, ripple reveals, and its orientation is selected one particular direction. In this chapter, we study ripple formation on Au(001) by single ion-beam sputtering. We first focus on patterning; how one can obtain well-defined ripple by controlling ion-beam flux, fluence, energy, and sample direction. Prior to review our results, we take a look at general features of so called sputter rippling on crystalline surface.

2.1 Determination of ripple orientation on crystalline surface

As we previously mentioned, the orientation of ripple is well described by the BH theory. However the BH theory assumed that surface is amorphous, not crystalline. This means it accounts for isotropic surface diffusion only. Thus, when we consider crystal surface, the BH theory could not work for ripple orientation because

underlying crystalline structure of surface affect surface diffusion as a directional bias, and thus morphology. Valbusa *et al.* [15] categorized two regimes according to whether ripple orientation follows ion beam or crystallographic direction. We call the former ‘erosive’ regime, the latter ‘diffusive’ regime.

2.2 Pattern formation on Au(001)

2.2.1 Experimental

All the experiments are carried out in ultra-high vacuum chamber. The base pressure is low 10^{-9} Torr. The initial clean Au(001) is prepared by series of Ar^+ sputtering and annealing up to 730 K. The incidence angle of ion beam from the surface normal is 73° . The ion energy, ϵ uses 0.5 keV or 2.0 keV. The flux, f ranges from about 6.0 to $13.0 \mu\text{A}/\text{cm}^2$. The sample temperature is initially room temperature, however it has slight increase by less than 20 K during the sputtering due to thermal accumulation by ion currents. The evolution of surface morphology is observed *ex-situ* by an atomic force microscope (AFM, PSI). All the images are AFM topographies, and ion beam projection direction is fixed as an arrow shown in Fig. 2.2.

2.2.2 SRIM results

For a given experimental condition, we can obtain ion characteristic lengths such as (a, σ, μ) from the SRIM [70] calculation. SRIM (Stopping and Range of Ions in Matter) is a popular software based on the TRIM (TRansport of Ions in Matter) code to calculate the stopping and range of ions (ranging from 10 eV up to 2 GeV/amu) into matter. TRIM is a Monte-Carlo algorithm introduced by Ziegler *et al.* and founded on the model of binary collision approximation for collision processes in solids. While it is most widely used for simulating ion-material interactions in solids and well-established in its accuracy for various materials, there are some limitations [71] and thus SRIM is not fully reliable.

By putting our experimental conditions i.e. the incidence angle of ion, ion type and its kinetic energy, and substrate material (The binding energy of gold is given with the default value 3.8 eV), into SRIM calculation, we obtain average ion penetration depth and its longitudinal and transverse ranges (widths). This is summarized in table 2.1.

Table 2.1. Ion characteristic lengths corresponding to our experiments are obtained by SRIM calculation.

Regime	θ	Energy (keV)	a (Å)	σ (Å)	μ (Å)
I (diffusive)	73	0.5	12	9	6
II (erosive)	73	2.0	23	18	13

We note that the coordinate system in SRIM for calculation of ion ranges is different from that introduced in the Sigmund theory (Only for the case of normal incidence of ions, it is same). Thus one should project the SRIM results onto the coordinate system corresponding to the Sigmund theory to obtain correct values of ion characteristic lengths.

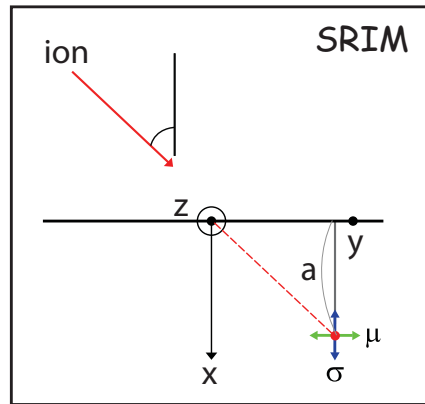


Figure 2.1. Coordinate system in SRIM calculation.

2.2.3 Image analysis

For the quantitative analysis of the pattern, we have calculated the two-dimensional height-height autocorrelation function $G(\mathbf{d})$

$$G(\mathbf{d}) = \langle h(\mathbf{r})h(\mathbf{r} + \mathbf{d}) \rangle, \quad (2.1)$$

where $h(\mathbf{r})$ is the relative surface height defined by $h(\mathbf{r}) - \bar{h}(\mathbf{r})$ where $\bar{h}$ is either mean height or local height in the best-fit plane calculated by raw data (to correct instrument-induced overall shapes). $\langle \rangle$ denotes spatial average over all pairs departing from $\mathbf{d}$. Its discretized version can be written by

$$\begin{aligned} G(i, j) &= \frac{1}{N^2} \sum_{n,m} [h(i+n, j+m) - \bar{h}(i+n, j+m)][h(n, m) - \bar{h}(n, m)] \\ &\approx \frac{1}{(N-i)(N-j)} \times \\ &\quad \sum_{n=1}^{N-i} \sum_{m=1}^{N-j} [h(i+n, j+m) - \bar{h}(i+n, j+m)][h(n, m) - \bar{h}(n, m)], \end{aligned} \quad (2.2)$$

where the surface topography is composed of N by N pixels. The (mean) ripple wavelength is determined from the one-dimensional autocorrelation function along the wave vector in which periodicity of the ripple is represented by well-defined satellite peaks. The separation between the central peak and the secondary peak gives the ripple wavelength λ .

2.2.4 Diffusive regime

Under ion flux $f \simeq 13.0 \mu\text{A}/\text{cm}^2$ and energy $\epsilon = 0.5 \text{ keV}$, a quasi ripple structure develops as shown in Fig. 2.2. Ridges enclose a pit, and are mostly aligned along the two principal surface directions, $[110]$ and $[1\bar{1}0]$ directions. In this case, the angle between $[110]$ direction and the ion-beam projection on the target surface is about 30° . Since the pattern is largely oriented along the $[110]$ direction not ion-beam

direction we regard this regime as diffusive regime. The orientation of the pattern is determined by $[110]$ direction in average sense, along which ridges are further elongated than $[1\bar{1}0]$ direction. The ripple pattern in Fig. 2.2 provides insight about nature of diffusion process in diffusive regime: (1) The orientation of sputter-induced pattern is determined by principal crystallographic direction, closer with the incident ion-beam direction because it brings longer ridge in average. Thus, orientation of pattern is not fully independent on the ion-beam direction even though we call it diffusive regime. (2) Role of surface diffusion in patterning is clearly shown; ion sputtering generates both adatoms and vacancies (which is definite from Molecular Dynamics simulation point of view [50]) on the surface, and supplies energy for them to promote diffusion. In this low flux (corresponding to high temperature since it allows longer time for adatoms/vacancies to diffuse), adatoms could favor to diffuse along with closely-packed step directions, $[110]$ or $[1\bar{1}0]$ direction, rather than fill vacancies due to step-edge barrier, the Ehrlich-Schwoebel (ES) barrier. Even though $[110]$ and $[1\bar{1}0]$ directions are equivalent, length of ridges along $[1\bar{1}0]$ direction is shorter than $[110]$ direction. The discrepancy definitely comes from in-plane ‘anisotropy’ of surface reconstruction by ion bombardment; when the incidence of ion beam is off-normal, the angular distribution of sputtered atoms change to a directional one along the ion-beam direction (in most samples), termed as ‘lateral sputtering’ [51].

2.2.5 Erosive regime

Much more flux (about 2 times than that in diffusive regime) and energy (4 times) of ion-beam conditions change the regime to erosion one (Fig. 2.3). The ripple is oriented along the ion-beam direction, not $[110]$ direction that differed from about 20° . Here, rather well-defined ripple is presented. Its characteristic length, wavelength of the ripple is determined from a line profile of 2D height-height correlation (autocorrelation) function across the ripple [Fig 2(b)]. The distance between main

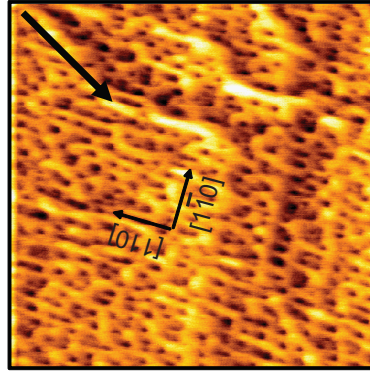


Figure 2.2. Topography observed after sputtering of Au(001) under the condition of diffusive regime. The arrow indicates ion beam direction which is deviated by about 30° from $[110]$ direction.

peak at center and next largest peak is defined as the ripple wavelength.

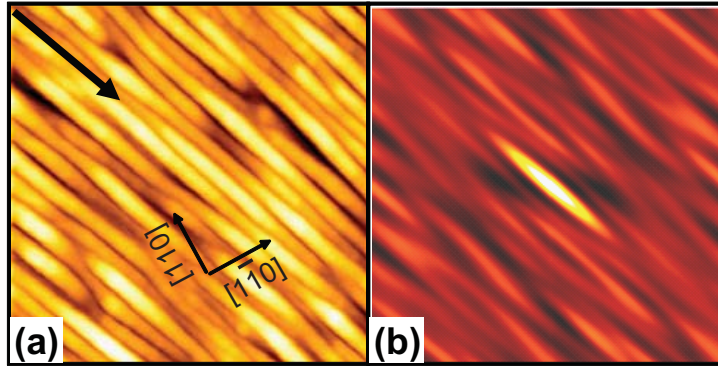


Figure 2.3. (a) Topography observed after sputtering of Au(001) under the condition of erosive regime. The arrow indicates incident beam direction where its angle deviation from $[110]$ direction is about 20° . (b) 2D height-height correlation function of (a).

2.2.6 Controlling ripple formation

Since the pattern formation by ion beam sputtering (IBS) relies on interplay between erosion by ions and smoothing by surface diffusion of ad-species (adatoms and vacancies), one can control them by means of ion fluence, flux, energy, and direction

with respect to crystallographic direction of the sample .

Fluence dependence

Figure 2.4(a,c) shows a ripple in diffusive regime where it is oriented along $[110]$ direction. In particular, along the projection direction of ions, troughs between ridges are extended and deeper more and more. Some areas are covered with holes due to crossing (along $[1\bar{1}0]$ direction) diffusion to the ion-beam direction. Varying only fluence double with respect to that of (a), the structure is enlarged. The wavelength is increased by about 30%, 30 nm from 23nm in (a). It shows like self similar ‘growth’ in deposition experiments. The detailed morphological features are still maintained; relatively long ridges along $[110]$ direction and irregular ridges bridging them along $[1\bar{1}0]$ direction.

Flux dependence

Next, to find better-ordered ripple structure, we control ion-beam flux. Fig. 2.5 shows the effect of flux on ripple structure. Increased by about 34% of ion-beam flux from $9.7 \mu\text{A}/\text{cm}^2$ (Fig. 2.5a) to $13.0 \mu\text{A}/\text{cm}^2$ (Fig. 2.5c), order of the ripple structure remarkably improved. Some areas, covered with hole-like structures in Fig. 2.5(a), are disappeared in Fig. 2.5(c). The coherence of ripple, i.e. the elongation length of ridge is extended over the scan area, which means larger than 500 nm.

Substrate orientation dependence

Moreover, we investigate dependence of substrate orientation with respect to the direction of the ion beam projection because the sample surface is crystalline. The Au(001) has two equivalent high symmetric axis, $[110]$ and $[1\bar{1}0]$, along which atoms are closely packed so that diffusion would occur more frequently. We check two cases; (1) The $[110]$ direction is matched with ion beam direction. (2) The

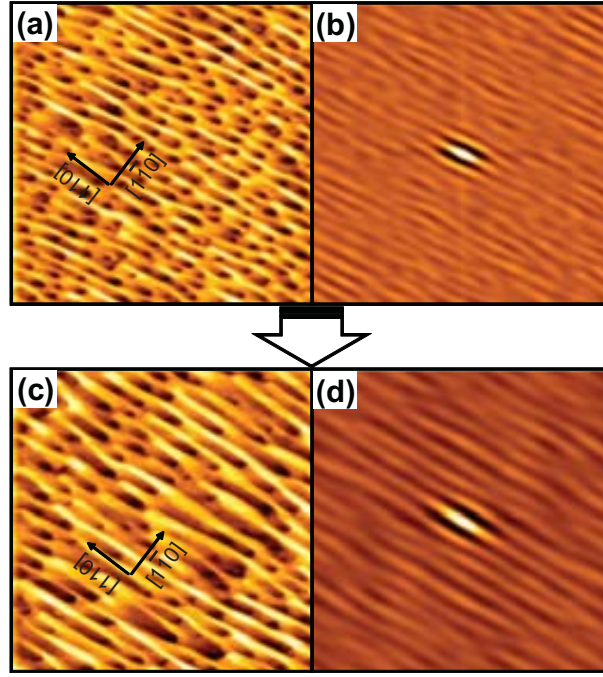


Figure 2.4. Fluence dependence of surface morphology. All conditions ($\epsilon = 2.0$ keV) are same except ion fluence. (a) $\Psi = 0.23 \times 10^4$ /nm². (c) $\Psi = 0.46 \times 10^4$ /nm². (b) and (d) are 2D height-height correlation function of (a) and (c), respectively. The scan area is 500×500 nm².

[110] direction is mismatched with ion beam direction. We refer to the former case on-axis sputtering and the latter case off-axis sputtering for simplicity. Figure 2.6 shows the effect of on/off axisness between ion-beam direction and substrate orientation. Even the same sputter condition such as $\epsilon = 2.0$ keV, $f = 13.0$ $\mu\text{A}/\text{cm}^2$, and $\Psi = 0.54 \times 10^4$ /nm², the surface morphology is drastically changed. When off-axis sputtering performed, the coherency of the ripple is clearly reduced by increased ‘branching’ of ripples [Fig. 2.6(c)]. From Fig. 2.6(a) one can determine this regime belongs to erosive regime because well-defined ripple is formed and aligned well with the ion beam direction. However, by rotating substrate slightly about 21° , the spatial order is degraded by diffusion process. Also it comes increasing roughness and wavelength.

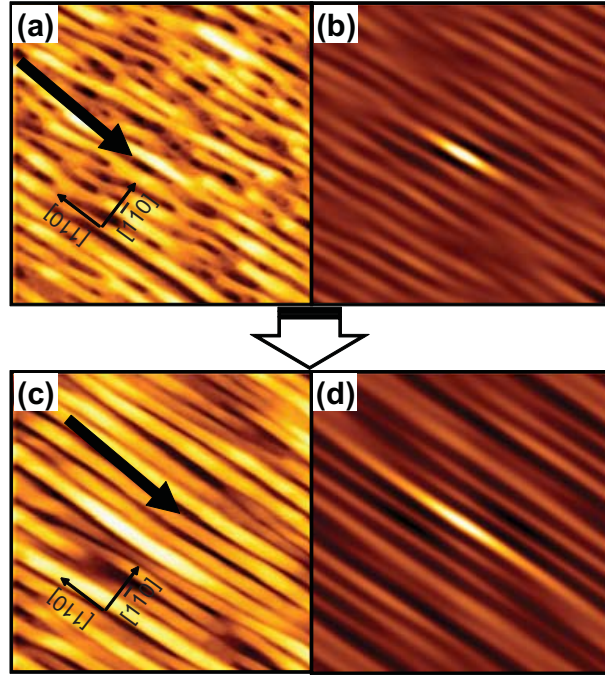


Figure 2.5. Flux dependence of surface morphology. All other conditions ($\epsilon = 2.0$ keV) are same except ion-beam flux (In fact, the ion fluence is slightly different; $\Psi = 0.44 \times 10^4$ /nm² for (a) and $\Psi = 0.54 \times 10^4$ /nm² for (c). However, it can be negligible error when we consider fluence dependence of pattern ordering because the fluence only participate in the pattern growth as shown in Fig. 2.4). (a) $f = 9.7$ $\mu\text{A}/\text{cm}^2$. (c) $f = 13.0$ $\mu\text{A}/\text{cm}^2$. Both arrows indicating ion projection coincide with [110] direction. (b) and (d) are 2D height-height correlation function of (a) and (c), respectively. The scan area is 500×500 nm².

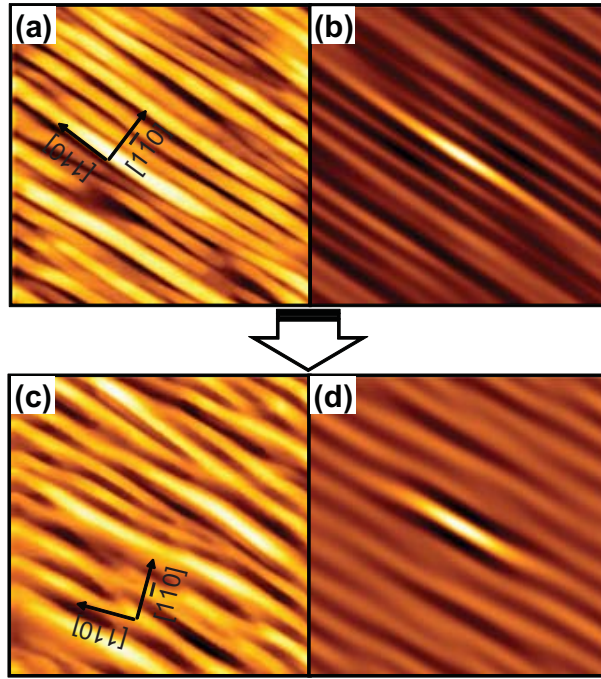


Figure 2.6. Dependence of surface morphology on substrate orientation. (a) On-axis sputtering case i.e. where the ion-beam projection lies on the crystal axis $[110]$ side by side. (c) Off-axis sputtering case where the angle difference between ion-beam direction and $[110]$ direction is about 21° . All other sputtering conditions are same ($\epsilon = 2.0$ keV, $f = 13.0$ $\mu\text{A}/\text{cm}^2$, $\Psi = 0.54 \times 10^4$ $/\text{nm}^2$). (b) and (d) are 2D height-height correlation function of (a) and (c), respectively. The scan area is 500×500 nm^2 .

To confirm our conjecture in off-axis sputtering, we check flux dependence of surface morphology in off-axis sputtering. By increasing flux (flux has role of inverse temperature because when flux is increased, atoms have less chance or time to diffuse.), one can reduce diffusion effect, however enforce erosion effect at the same time. Figure 2.7(a,b) looks like checker board which consists of nanoholes even though slight elongation along $[110]$ direction of ridges are displayed. As the flux is doubled [Fig. 2.7(d,e)], the pattern is changed remarkably more ripple-like. However it still shows incomplete ripple by many branches. The ripple direction is more closed to ion-beam direction, but rigorously speaking, in-between ion beam direction and the crystalline axis, $[110]$ direction.

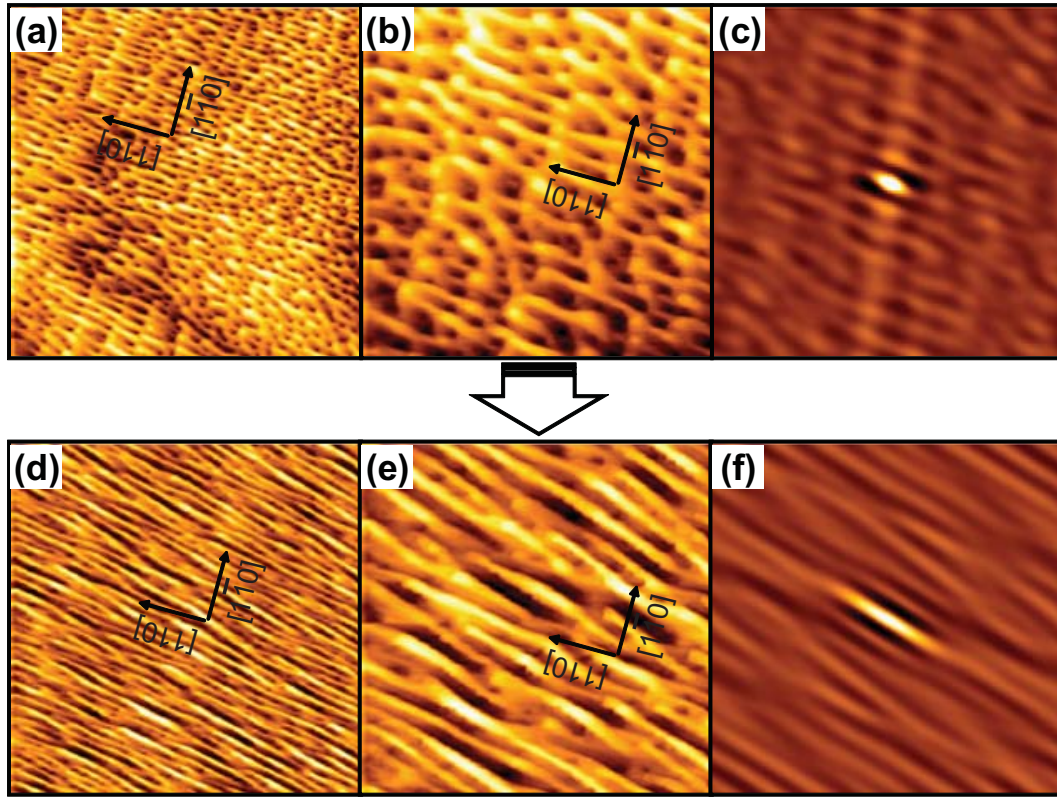


Figure 2.7. Flux dependence of surface morphology in off-axis sputtering. (a) Off-axis sputtering case with relatively low flux, $f = 6.3 \mu\text{A}/\text{cm}^2$. (d) A case with higher flux, $f = 13.0 \mu\text{A}/\text{cm}^2$. All other sputtering conditions are same ($\epsilon = 2.0 \text{ keV}$, $\Psi = 0.24 \times 10^4 / \text{nm}^2$). (b) and (e) ($500 \times 500 \text{ nm}^2$) are zoomed image of (a) and (d) ($1 \times 1 \mu\text{m}^2$), respectively. (c) and (f) are 2D height-height correlation function of (b) and (e), respectively.

Energy dependence

Finally, we check energy dependence of ripple formation in the diffusive regime. The only difference between Fig. 2.8(a,b) and (c,d) are the ion energy 0.5 keV and 2.0 keV, respectively. As the ion energy become four times, more rough surface is obtained; RMS roughness is changed from 9.5 Å [Fig. 2.8(a)] to 16.8 Å [Fig. 2.8(c)]. The wavelength is also increased with the energy being contrary with Eq. (1.21). It may come from energy dependence of a/σ or a/μ , which is negligible in most cases. By aid of SRIM simulation, one can easily obtain the energy dependence of the ion characteristic parameters such as a , σ , and μ . Ion energy dependence on ripple wavelength has been studied on usually non-metal targets [5, 6, 59], except recent study on polycrystal Al surface [60]. If we compare the morphology change with that of Fig. 2.7, where flux is doubled, flux is more crucial factor than ion energy for increasing ripple coherency.

2.3 Summary and Discussion

Controlled patterning of a ripple structure by tuning sputter parameters is investigated. A sufficiently large flux and alignment of the crystal axis of the substrate to the in-plane ion beam direction are required to overcome the degradation of ripple, mainly due to the surface diffusion. The ion energy and fluence influence the growth of amplitude, but not the order or shape of the pattern; the quantities such as roughness and wavelength can be controlled by adjusting ion fluence and energy. These are summarized in Table 2.2. Under our sputter condition, the ripple orientation represents well the competition between ion beam direction and crystallographic direction. For instance, at an intermediate regime, effects of both are clearly visible in terms of the degradation of ripple ordering, by alignment along a direction between the ion beam projection and the high symmetry crystalline axis. Compared to the Bradley-Harper theory, which assume an amorphous substrate and thereby could

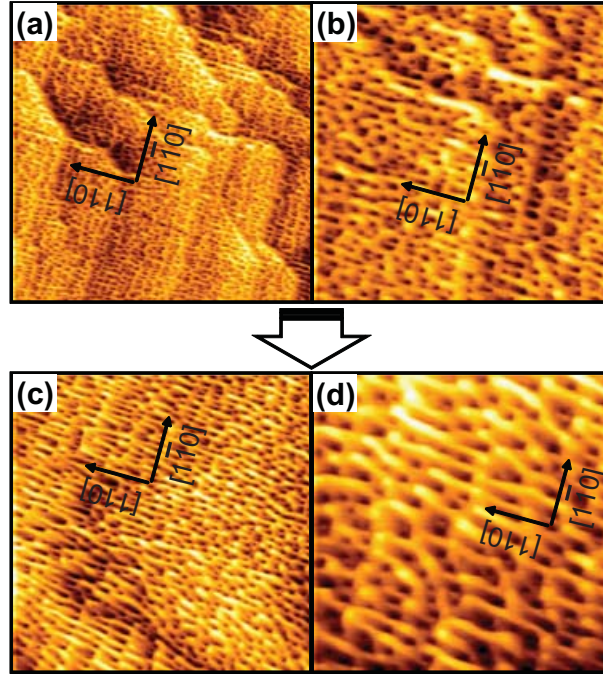


Figure 2.8. Energy dependence of surface morphology in diffusive regime. Only ion energy differs; $\epsilon = 0.5$ keV for (a) and $\epsilon = 2.0$ keV for (c). Image size: (a, c) $1 \times 1 \mu\text{m}^2$; (b, d) $500 \times 500 \text{ nm}^2$. Other sputtering conditions are as follows: $f = 6.3 \mu\text{A}/\text{cm}^2$ and $\Psi = 0.24 \times 10^4 \text{ /nm}^2$.

allow only one orientation at a grazing angle, the effect of the anisotropic diffusion in this crystalline metal substrate causes more rich features e.g. the marginally oriented ripple [Fig. 2.6(c,d)] and checkerboard-like ripple in diffusive regime.

Valbusa obtained a rich surface pattern on crystalline metal surfaces by control of temperature [15, 57], where the temperature has the role of activating surface diffusion. According to his description, there is a hierarchy in activation energy (for terrace diffusion) and the Ehrlich-Schwoebel (ES) barrier for surface diffusion; the densely-packed direction $[110]$ makes it easier for the atoms to hop, and the energy barrier is lower than $[001]$ direction. Thus at certain temperatures surface diffusion along $[110]$ direction is significant, but still minor in $[001]$ direction. This anisotropy in surface diffusion reflects into the surface morphology. ES barriers

have also anisotropy (simply counting bonding numbers gives such directional dependence.) and the same manner of description is applied.

Instead of varying temperature, we controlled the surface patterns by means of flux. The flux J is known to have a role of inverse temperature T effectively. Because if flux i.e. the ion impingement rate is large, the surface atoms do not allow much time to diffuse. Rather, it has enough time to diffuse under low flux. This is well applied to our results. We obtained a seemingly continuous transition from diffusive regime to erosive regime by increasing flux at room temperature. Our results confirm the interplay between surface diffusion kinetics and the ion-induced kinetics is crucial to develop final morphology, especially on crystalline metal substrate.

We note that the ‘inverse’ relationship between flux and temperature should not be taken to mean literally $J \sim 1/T$; if J increases, the incoming energy increases, and thus the substrate temperature T would be increased. That is just a qualitative description, and it should be understood according to Ref. [3]: the ratio of D/J i.e. ion-beam deposition (which induced erosion) to diffusion rate is the essential parameter characterizing growth kinetics, where D is the diffusivity, which obeys Arrhenius law (Eq. 1.6). If the ratio D/J is large, the surface morphology is mostly governed by surface diffusion i.e. it allows enough time for diffusing species to travel around the surface to reach a minimized energy configuration. If the ratio is small, the pattern formation is dominated by the ion-beam stimulated kinetics (erosion and redistribution of surface atoms), which result in various metastable structures like dot and ripple.

Since the geometry between ion-beam and target surface has a significant role for making better ordered ripple structure in this broad ion-beam sputtering technique, we propose the following idea to improve the spatial order of ripple structure further: Incident ions from a common commercial ion gun have distribution in its angle. The smaller the grid radius of the ion gun becomes, the larger the angular

deviation from a source would be. Thus if we use an ion gun having a large radius of grid, so that nearly perfect uniform ions are reached on the target surface, the quality of ripple pattern (the ordering and its coherency) would be further enhanced [54].

Table 2.2. Dependence (qualitative description) of surface morphology of Au(001) on various sputter conditions.

Control	Ψ	ϵ	f	On-axisness
Order/Shape	Minor	Minor	<i>Crucial</i>	Major
Size (W , A , and λ)	Major	Major	Minor	Minor

Chapter 3

Nanopatterning by dual-ion-beam sputtering

We studied development of ordered nanopatterns by employing dual-ion-beam sputtering (DIBS). In DIBS, two ion beams, crossing perpendicular to each other at their azimuth, are simultaneously incident on Au(001) at a grazing angle. In the erosion (diffusion) regime, a square-symmetric two-dimensional (2D) pattern of nanodots (holes) is formed. The 2D pattern is achieved only when the two ion-beam fluxes are well- balanced in the erosive regime. While the diffusive regime does not require such strict condition. These findings were not properly explained by the derived Kuramoto–Sivashinsky (KS) equation, responsible for DIBS.

3.1 Introduction

Recently, ion-beam sputtering has been widely applied to grow self-organized nanostructures on various substrates, ranging from insulators and semiconductors to metals. For example, a hexagonal pattern of nanodots was formed on a GaSb surface by ion-beam sputtering at normal incidence [4]. If the main surface species are vacancies, nanohole arrays are observed [57,58]. In contrast, at an off-normal incidence, periodic ripples can be produced on the irradiated surfaces [6, 12, 14]. Thus far,

such sputter-induced patterning has been performed using a single ion beam, which might limit the variety of nanopatterns. To achieve the goal of nano-sculpturing a surface as designed, we are truly in need of removing such constraint.

Carter [45] proposed the use of dual-ion-beam sputtering (DIBS) to produce more versatile nanostructures. During DIBS, the two beams are expected to induce interference effects and produce sophisticated patterns. Recently, sputter-induced patterning using multiple ion beams has been studied [46]. However, this was a numerical study, and no experimental study has examined the idea of patterning by multiple ion-beam sputtering.

we report the experimental realization of DIBS, in which two ion beams are simultaneously incident on Au(001) at a grazing angle and perpendicular to each other at their azimuth (Fig. 3.1a). As a result of DIBS, we found two dimensional patterns of nanoholes (Fig. 2) and nanodots (Fig. 3) instead of ripples, although sputtering is carried out at grazing angle with a sample at rest [61]. In the diffusive regime where the energy and flux are relatively weak and the pattern formation is largely governed by the diffusion of the main surface species, square-symmetric pattern of nanoholes is formed where the edges of the squares are aligned along the high symmetric directions of the substrate. In the erosive regime, beam energy and beam flux are large enough to let the erosion be the dominant effect in the patterning process. In this case, the pattern orientation is independent of the crystallographic directions of the substrate, and found along the ion beam directions. In the erosive regime, the formation of the nanodot pattern is highly sensitive to the balance of the two crossing ion beams. Otherwise, a ripple pattern with modulated height prevails (Fig. 4). This is not the case in the diffusive regime, possibly due to the effective diffusion kinetics of adatoms and/or vacancies that cure the defects formed by ion beams so as to keep the shape of ripples uninterrupted.

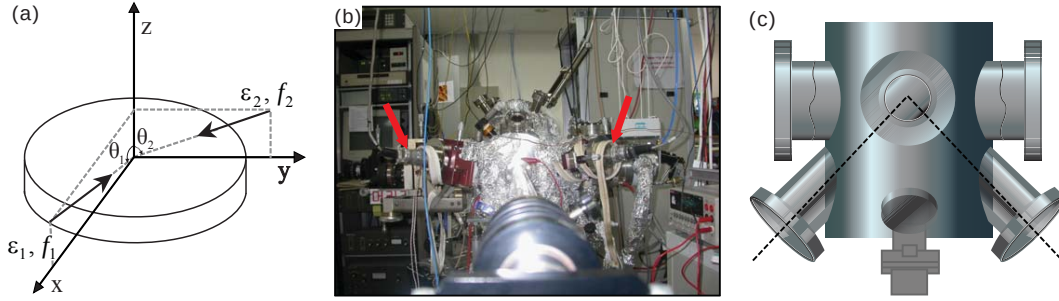


Figure 3.1. (a) Schematic illustration of the experimental geometry. (b) The DIBS chamber; red arrows indicate two ion guns. (c) Top view of DIBS chamber (illustration). Ions from two ion guns arrive the center of target surface along dashed lines, which exerted from each ion-gun port.

3.2 Experimental

All the sputtering experiments were performed in a custom-built ultrahigh vacuum chamber. The base pressure of the chamber is in the range of mid to high 10^{-10} Torr. Clean and atomically flat Au(001) surfaces are prepared by a series of Ar⁺ sputtering and annealing cycles with annealing temperature of 730 K. The mean terrace width of the clean Au(001) is of the order of several hundred nanometers.

During DIBS, the impinging ion beams make a grazing angle of $\theta \approx 73^\circ$ to the surface normal. (Fig. 3.1) Each ion gun is mounted on a goniometer that allows the adjustment of incident angle in two directions perpendicular to each other. The ion energy ϵ and ion flux f for each ion beam are operated in two modes: (i) $\epsilon = 0.5$ keV and f less than 1 ion/nm²s and (ii) $\epsilon = 2.0$ keV and f larger than 1 ion/nm²s [62]. The former case corresponds to the diffusive regime, while the latter to the erosive regime [63]. Slight increase of sample temperature from the initial temperature (room temperature) by less than 20 K was observed at the last stage of sputtering due to thermal accumulation as a result of extended sputtering. The morphological evolution of Au(001) by ion beam sputtering was investigated *ex situ* by atomic force microscopy in contact mode.

3.3 DIBS in the diffusive regime

Figure 3.2 shows the development of nanopatterns on Au(001) by DIBS in the order of increasing ion fluence $\Psi (= f \times t)$. Fig. 3.2(a) shows a representative, initial Au(001) displaying wide terraces of several hundred nanometer in their average width. For $\Psi \approx 5.2 \times 10^3/\text{nm}^2$, ordered holes are clearly visible in Fig. 3.2(b). They are rather closely packed and have four-fold symmetry with edges along [110] directions (the inset of Fig. 3.2(b)). The inset shows the height-height correlation function $G(\mathbf{d}) = \langle h(\mathbf{r})h(\mathbf{r} + \mathbf{d}) \rangle$, where $\langle \dots \rangle$ denotes the spatial average and the vector $\mathbf{d}$ does displacement from a given position $\mathbf{r}$. Each hole is enclosed by four, nearly straight ridges, suggesting that the pattern originates from crossing ripples. The ridges are, however, not aligned along the beam directions marked as arrows in Fig. 3.2(a), but along the densely packed crystallographic directions, [110]. This indicates that the nanopatterning is governed by diffusion of surface species. As the ion fluence increases further (Fig. 3.2(c,d)) lateral and vertical growth of the holes is observed, while the symmetry and order of the nanopatterns gradually degrades. This could be due to the build up of thermal agitation. Fig. 3.2(e) shows the wider area image obtained by a scanning electron microscope. It confirms the formation of nano-hole pattern, and its uniformity and long range order can be seen more clearly in this image.

Such nano-hole arrays have been already reported for different materials [64, 65]. However, all of them show relatively poor lateral order, because nano-hole arrays form in diffusive regime by thermally activated vacancies. The relatively high order of the present nano-hole pattern may be attributed to the preferential diffusion of vacancies along ripples that offer highly efficient diffusion channels along their step edges. To improve the order of sputtered pattern, sputtering of pre-patterned surface have been explored, recently [84]. From that perspective, during DIBS, we witness self-templated (by the ripples), and self-organized growth of nano-holes.

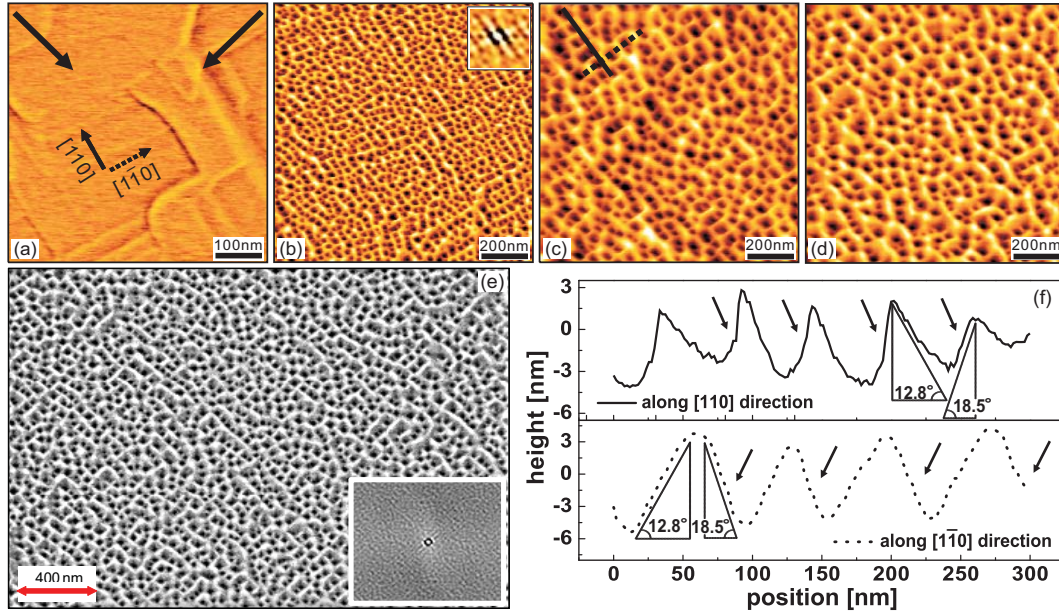


Figure 3.2. (Color online) (a) Initial Au(001) sample. (b)-(d) Nanopatterns induced by DIBS under the diffusive regime, $\epsilon = 0.5\text{keV}$ and $f = f_1 + f_2 = 0.93 \text{ ions/nm}^2\text{s}$. They are presented in the order of increasing sputter time. (b) $\Psi = 5.2 \times 10^3/\text{nm}^2$ and (c) $\Psi = 2.0 \times 10^4/\text{nm}^2$. (d) $\Psi = 3.5 \times 10^4/\text{nm}^2$. (e) A representative SEM image taken at $\Psi = 2.0 \times 10^4/\text{nm}^2$ and its corresponding autocorrelation function in the inset. (f) Height profiles along the two lines marked in (c). The arrows indicate the incident ion-beam directions.

Fig. 3.2(f) shows representative line profiles along the two crystallographic directions as marked by a solid and a dashed lines in Fig. 3.2(c). Comparing the two line profiles, we find that the ridge along $[\bar{1}\bar{1}0]$ (solid line) are broader than those along $[110]$ (dashed line). The mean wavelength along $[\bar{1}\bar{1}0]$ and $[110]$ estimated from the height-height correlation function are $65 \pm 3\text{nm}$ and $54 \pm 2\text{nm}$, respectively. This reflects the difference between the two ion-beams employed for DIBS. Another distinguished feature is that the ridges show slope selection with different slopes on their two opposite faces (Fig. 3.2(f)); (1,1,4) facet is frequently found on the illuminated (by a crossing ion beam) side, while (1,1,6) facet is on the shadowed side. Similarly asymmetric slopes of ripples are also observed for ripples formed in the erosive regime (Fig. 3.3(e)). It is tempting to attribute this asymmetry to the slope dependence [34, 39] of the erosion rate by the ion beam crossing the ripple. However, the different erosion rates between the two sides, as far as it is uniform on each side, result only in different lateral erosion speeds of the two sides of ripples, while the slopes on both sides remain the same. This indicates that their should be effects which are dependent on incidence angle, besides the sputter erosion, that produce asymmetric slopes.

3.4 DIBS in the erosive regime

In the erosive regime, ordered patterns of nanodots develop as can seen in Fig. 3.3. Fig. 3.3(a) shows the incipient stage of nano-dot formation. There are found tiny dots of varying sizes and shapes. In some region they already show square symmetric order of the nano-dots. As Ψ increases, the dots grow and become more easily discernable, but most of them are still linked with each other (Fig. 3.3(b)). In Fig. 3.3(c) openings are found sporadically, and the present pattern closely resembles a labyrinth (Fig. 3.3(c)). After further sputtering, a pattern of separated dots is formed (Fig. 3.3(d)). They are fairly well ordered, and show square-symmetric

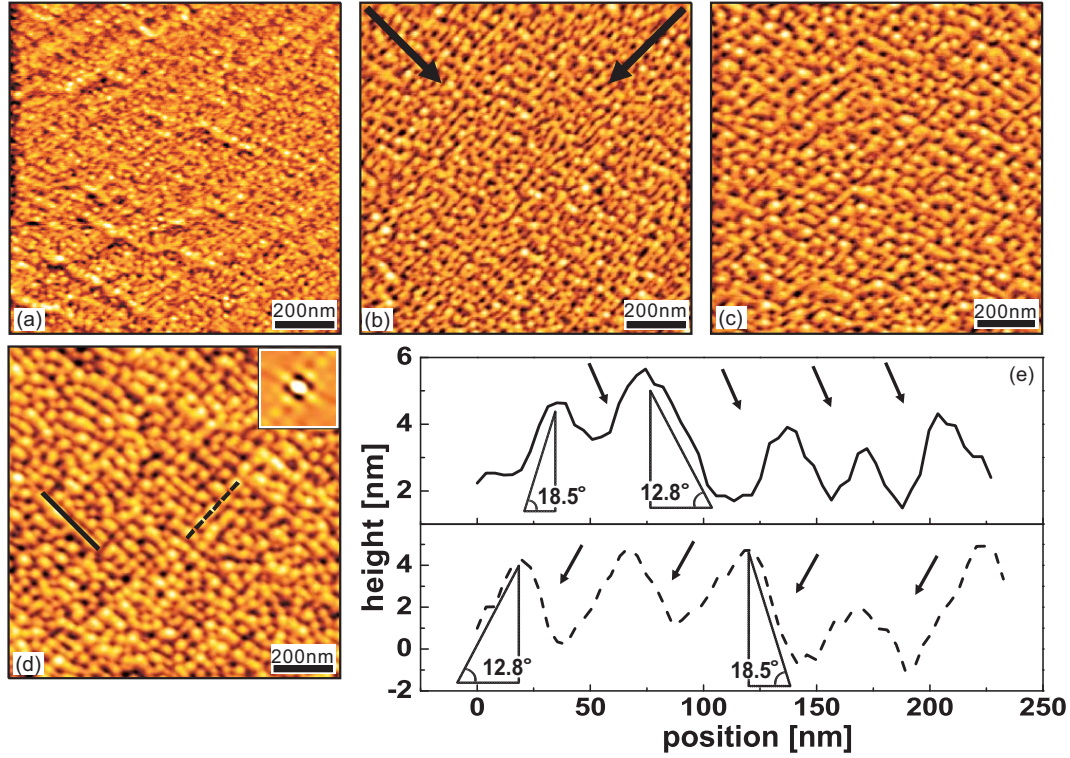


Figure 3.3. (Color online) Nanopatterns induced by DIBS under the erosive regime, $\epsilon = 2$ keV and $f \approx 3.25$ ions/nm²s. The figures (a)-(d) are presented in the order of increasing sputter time. $\Psi = 940/\text{nm}^2$ ($t = 5$ min), $\Psi = 1900/\text{nm}^2$ (10 min), $\Psi = 3900/\text{nm}^2$ (20 min) and $\Psi = 6350/\text{nm}^2$ (32 min) from (a) to (d), respectively. Inset in (d) is the height-height correlation function.

correlations, as shown in the height-height correlation in the inset. However, we clearly see that the sizes of the dots vary much, and there are many point defects that are more or less coordinated than four.

It is noteworthy that in the erosive regime, the 2-D pattern of the nanodots as seen in Fig. 3.3 is achieved only under the balance of the two ion beams. Otherwise modulated ripples as in Fig. 3.4(a) are observed. The 2-D nano-hole pattern, on the other hand, is formed in the diffusive regime with no strict requirements for the balance of the two ion beams. The modulated ripples are sharply contrasted with the ripples formed by single-ion-beam sputtering that show no modulation in height

and width (Fig. 3.4(b)). The observed modulation in Fig. 3.4(a) is reminiscent of the incomplete breakage of the ripples into dots by the crossing (minor) ion beam of less flux.

We can understand the two distinct behaviors under the two sputtering conditions as follows: The role of each beam is two-fold during DIBS. Each ion beam produces ripples along its direction, while it simultaneously destroys the ripples perpendicular to it. If the balance between the two beams is not met, then one beam (major beam) breaks a crossing ripple into nanodots, while the other beam (minor beam) is incapable of breaking it into separate nanodots and simply causes modulation in its crossing ripples. As a result, the nanodot pattern is observable only under very restrictive conditions where both beams break their crossing ripples into nanodots simultaneously. In the diffusive regime, however, the diffusion of the adatoms and/or vacancies along the edges of the ripples are so fast [63] that the defects in ripples caused by the crossing ion beam is efficiently cured, and the remaining ripples can form the ripple-bounded holes under most sputtering conditions, or the 1-D instability forming nano-dot array from a ripple is absent in the diffusive regime.

3.5 Superposition works?

It would be interesting to find whether the pattern induced by DIBS results from the superposition of the two crossing ripples, each of which is independently formed by respective ion beam employed for DIBS [45, 46]. To address this issue, we perform ripple formation by each ion beam independently under the same condition for DIBS, and then superpose the two ripple patterns digitally to compare it with that of DIBS.

Figs. 3.5(a) is a pattern obtained by DIBS under diffusive condition, while Fig. 3.5(b) is the digitally superposed image of the two, independently formed ripple patterns. Under the erosive regime, we obtain a pattern upon DIBS like Fig. 3.5(c)

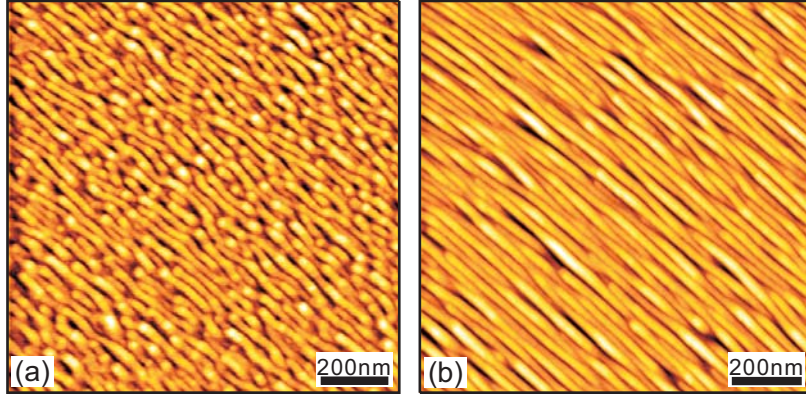


Figure 3.4. (Color online) Nanopatterns induced by DIBS with unbalanced dual beams (a) and single ion beam sputtering (b). For the case of DIBS, the beam fluxes are varied from that for Fig. 3.3(d); f_1 is decreased by about 5%, while f_2 is increased by about 30% with respect to the case of Fig. 3.3(d)). For the case of single ion beam sputtering, the beam flux is the same as f_2 of Fig. 4(a).

and a superposed image, Fig. 3.5(d). For both regimes, the superposed images do not reproduce those by DIBS. The real image show larger characteristic lengths of the nanostructures and their spacings than those of the superposed images. Moreover, in the diffusive regime, the nano-feature in the real image is nano-hole rather than nano-dot observed in the superposed image. In addition, the instability of the nano-dot pattern observed under the erosive regime is absent for the superposed pattern. These all combined look to evidence the inapplicability of superposition of ripples.

3.6 Effect of temperature

One may, however, wonder whether the increased size of the characteristic lengths in both regimes may be the consequence of the increase in the substrate temperature due to the doubled flux in the DIBS [66] in comparison with the case for single ion beam sputtering. In fact, we observe an increase by ≤ 20 °C in the sample temperature at the end of each sputtering term. According to the linear theory proposed by

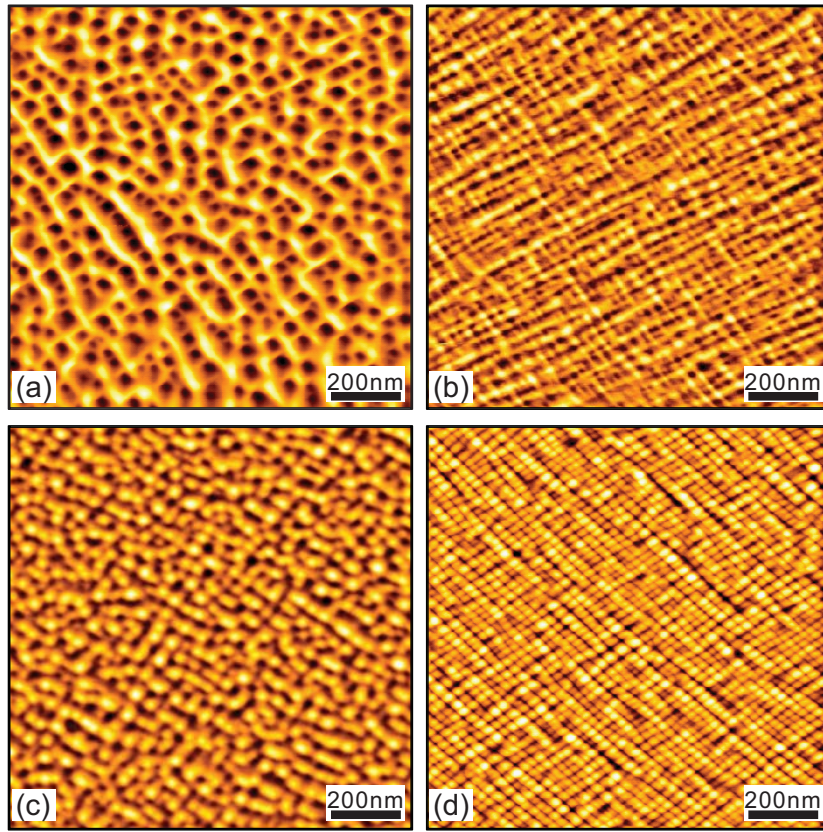


Figure 3.5. (Color online) Digitally superimposed images in the diffusive (a) and erosive regime (c). (b) and (d) are corresponding DIBS results; (b) for the diffusive regime, and (c) for the erosive regime.

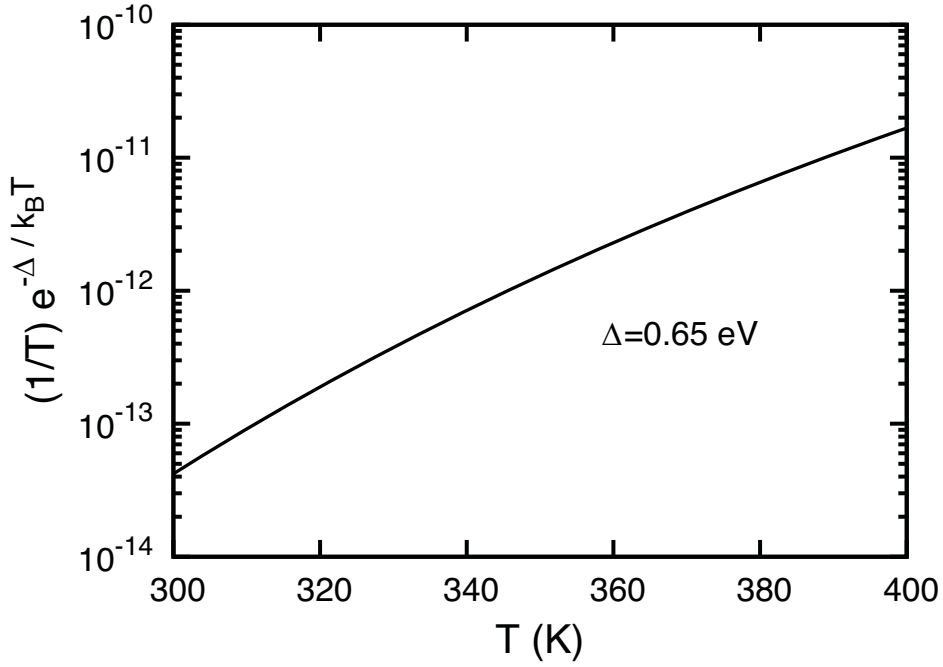


Figure 3.6. Semi-log plot of the Arrhenius law, $(1/T)\exp(-E(= 0.65\text{eV})/k_B T)$. As T increases, thermally activated diffusion coefficient K increases with T , thus wavelength predicted by BH also increases in nearly exponential manner.

Bradley and Harper [21], the wavelength of ripples follows

$$\ell = 2\pi \sqrt{\frac{2K}{|\nu|}} \propto \sqrt{K(T)} \propto \sqrt{\frac{1}{T} \exp(-E/k_B T)}, \quad (3.1)$$

where K is the thermal diffusion coefficient, which follows Arrhenius law at high temperature; E , the local energy barrier; k_B , the Boltzmann constant; and T , the substrate temperature. Indeed, by using the known terrace diffusion barrier on Au(001), $E \approx 0.65$ eV [85], we estimate the increase in the characteristic size to be approximately 112 % due to the increase of the substrate temperature by 20 °C. Thus, the enhanced temperature can be a cause for the discrepancy in the characteristic lengths between the DIBS pattern and the superposed ripple pattern.

Furthermore, as the substrate temperature increases, the species shaping surface may change from adatoms to vacancies, since the diffusion barrier for advacancy is

higher than that for adatom [57]. Actually, such change of surface structure from adatom islands to vacancy islands have been reported with the increase of substrate temperature [57, 66, 67]. The ion flux is lower for the diffusive regime than that for the erosive regime, and thus the effective temperature that approximately scales inversely to the flux [33] should be higher in the former than in the latter. Accordingly, the switch of surface species is more probable and actually observed for diffusive regime. In short, the thermal effects are so critical in shaping Au(001) during sputtering, while the present experiments have been performed with its temperature not fully stabilized. Further experiments with stringent control of substrate temperature is required to reach a solid conclusion whether the superposition of ripples work during DIBS.

Estimation of K

From data from Au(001), we can estimate the magnitude of thermal relaxation rate K . The surface diffusion constant D_0 in the Eq. (1.5) has following value for Au(001) [56]

$$D_0 \simeq 5.0 \times 10^{-3} \text{cm}^2/\text{s} = 5.0 \times 10^{13} \text{\AA}^2/\text{s}. \quad (3.2)$$

The surface tension γ is given by [74]

$$\gamma = 1.63 \text{J/m}^2 = 0.1 \text{eV/\AA}^2. \quad (3.3)$$

The areal atomic density n is estimated by 2 atoms in a_0^2 area, where $a_0 = 4.198 \text{\AA}$ from [74] i.e.

$$n = 0.11/\text{\AA}^2, \quad (3.4)$$

and atomic density N is given by

$$N = 5.9 \times 10^{22}/\text{cm}^3. \quad (3.5)$$

Putting above quantities into the Eq. (1.5), one can obtain

$$K = 7.7 \times 10^4 \text{\AA}^4/\text{s}, \quad (3.6)$$

where we assumed $T = 300$ K and $\Delta E = 0.65$ eV. For comparison, Si(001) at $T = 823$ K has $K \simeq 1.15 \times 10^6 \text{ \AA}^4/\text{s}$. From SRIM results 2.1 and Ref. [20], we can also estimate an order of magnitude for the ion-induced surface diffusion constant $D_x x$. By using Eq. (1.16),

$$F \simeq 0.41 \text{ \AA}/\text{s}. \quad (3.7)$$

To compare with K , one should multiply F by $D_x x = 2840 \text{ \AA}^3$ in table 3.1, which is a value divided by the global factor F . Thus, it becomes $1.16 \times 10^3 \text{ \AA}^4/\text{s}$, which is one order of magnitude smaller than K . It means that thermal surface diffusion is more relevant than ion induced smoothing at $T = 300$ K. We, however, still wonder that effective temperature under the erosive regime would be lower than $T = 300$ K, because the surface morphology is dominated by ion-beam erosion not by surface diffusion. We phenomenologically recognize that ion flux and effective temperature has inverse relation [63], however its specific quantitative description has been lacking.

3.7 Scaling behavior in DIBS

Surface roughening and wavelength coarsening are simultaneously presented under DIBS. The behavior has been understood by the scaling theory [24]. The surface roughness of the sample, having $L \times L$ in size, at sputter time t is defined as $W \equiv \sqrt{(1/L^2) \sum_{\mathbf{r}} [h(\mathbf{r}, t) - \bar{h}]^2}$, where $\bar{h} = (1/L^2) \sum_{\mathbf{r}} h(\mathbf{r}, t)$. For many deposition or erosion process, It is well known that a scaling relation works; $W(L, t) \sim L^\alpha g(t/L^{\alpha/\beta})$, where $g(u) \sim u^\beta$ for $u \ll 1$ and $g(u \rightarrow \infty) \sim \text{constant}$. The roughness exponent α and growth exponent β are related through the scaling relations, which gives the dynamic exponent, $z = \alpha/\beta$. It determines the scaling of the saturation time with the system size L . The inverse dynamic exponent, $1/z$ is called the coarsening exponent. Qualitatively speaking, surface looks more rough at small α because of many troughs and hills in smaller scale even though the

roughness is same.

To acquire further insight of the growth kinetics during DIBS [24], we measure the surface roughness at each fluence, and obtain $\beta = 0.35 \pm 0.06$ in the diffusive regime and $\beta = 0.47 \pm 0.06$ in the erosive regime. These values are larger than the universal value $\beta \approx 0.24$ for the Kadar-Parisi-Zhang equation and $\beta = 0.16 - 0.25$ for the KS equation in two dimensions [79], but supports the present argument that DIBS does not follow the KS equation. We note that our system has not reached stationary state even at the largest fluences reported. Furthermore, coarsening of nanostructures are clearly observed; wavelength (λ) increases from 37 – 43 nm in Fig. 1(c) to 52 – 68 nm in Fig. 1(d) in diffusive regime and from 37 – 45 nm in Fig. 2(a) to 43 – 48 nm in Fig. 2(b) in erosive regime.

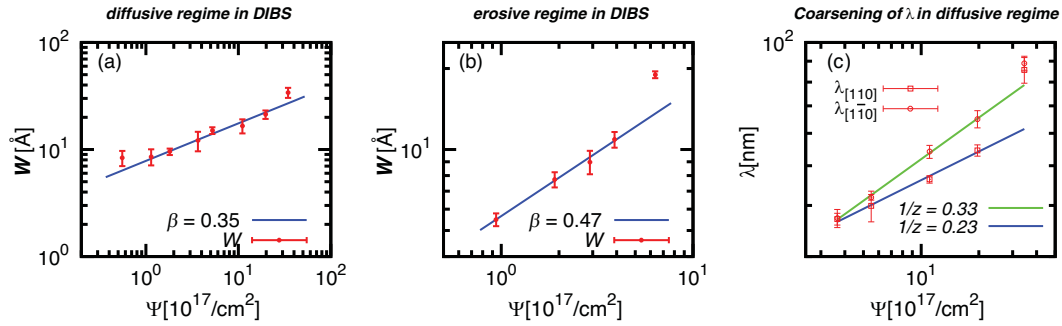


Figure 3.7. Ion fluence dependence of the surface roughness W (a,b) and the ripple wavelength λ (c) under DIBS. Three points at early times are missed in (c) because the pattern is too small to identify.

3.8 Continuum-level description for DIBS

We derive a continuum equation for DIBS starting from the Sigmund theory in which we add a term to account for the energy contribution from another ion-beam in DIBS. The derived equation has reduced to the same form as the KS equation (1.15); however, the coefficients are changed.

The first attempt at a theoretical approach for DIBS would be to set up a continuum equation taking the simultaneous incidence of the two beams. For this purpose, we follow the derivation of the KS equation for SIBS [20]. For the case of dual ion beams, the contribution of the another beam to erosion velocity at ‘ o ’ V_o can be simply treated as

$$V_o^{\text{DIBS}} \propto \int d\mathbf{r} \{ \varepsilon_1(\mathbf{r}) f_1(\mathbf{r}) + \varepsilon_2(\mathbf{r}) f_2(\mathbf{r}) \}. \quad (3.8)$$

Here, any interference effect between the two ion-beam fluxes is not taken into account. One might think it is too simplified and then could suggest following equation in which appear cross terms correlated between two beams’ flux and energy:

$$V_o^{\text{DIBS}} \propto \int d\mathbf{r} \{ f_1(\mathbf{r}) + f_2(\mathbf{r}) \} \{ \varepsilon_1(\mathbf{r}) + \varepsilon_2(\mathbf{r}) \}. \quad (3.9)$$

However, this is not correct. Think about a case; sputtering at normal incidence with 1 keV Ar^+ ions and average flux J . It is identical with the situation where two fluxes, each $J/2$ with 1 keV Ar^+ ions bombard a target. In contrast, a case such as two fluxes $J/2$ with 0.5 keV Ar^+ ions is physically different with the original situation. The total energy input is a half of that of the original case at a given time. But the Eq. (3.9) same result with the original. Thus the Eq. (3.8) is relevant. The flux can be separated but the energy can not. We think flux and corresponding energy should be product by a pair to avoid such strangeness.

We can separate integral in Eq. (3.8) in terms of each flux and thus finally reach a superposed KS equation in which coefficients are obtained from the respective KS equation corresponding to each beam. This is carried on by coordinate transformation with rotation matrix. For example,

$$\begin{pmatrix} \frac{\partial h}{\partial x_2} \hat{x}_2 \\ \frac{\partial h}{\partial y_2} \hat{y}_2 \end{pmatrix} = \begin{pmatrix} \frac{\partial x_1}{\partial x_2} = \cos \varphi & \frac{\partial y_1}{\partial x_2} = \sin \varphi \\ \frac{\partial x_1}{\partial y_2} = -\sin \varphi & \frac{\partial y_1}{\partial y_2} = \cos \varphi \end{pmatrix} \begin{pmatrix} \frac{\partial h}{\partial x_1} \hat{x}_1 \\ \frac{\partial h}{\partial y_1} \hat{y}_1 \end{pmatrix},$$

and by using this one can replace high order derivative terms with those for x_1 - y_1 coordinate as follows:.

$$\begin{aligned} \frac{\partial^2 h}{\partial x_2^2} &= \frac{\partial}{\partial x_2} \left(\cos \varphi \frac{\partial h}{\partial x_1} \hat{x}_1 + \sin \varphi \frac{\partial h}{\partial y_1} \hat{y}_1 \right) \\ &= \left(\cos \varphi \frac{\partial}{\partial x_1} \hat{x}_1 + \sin \varphi \frac{\partial}{\partial y_1} \hat{y}_1 \right) \left(\cos \varphi \frac{\partial h}{\partial x_1} \hat{x}_1 + \sin \varphi \frac{\partial h}{\partial y_1} \hat{y}_1 \right) \\ &= \cos^2 \varphi \frac{\partial^2 h}{\partial x_1^2} + 2 \cos \varphi \sin \varphi \frac{\partial^2 h}{\partial x_1 \partial y_1} + \sin^2 \varphi \frac{\partial^2 h}{\partial y_1^2}. \end{aligned} \quad (3.10)$$

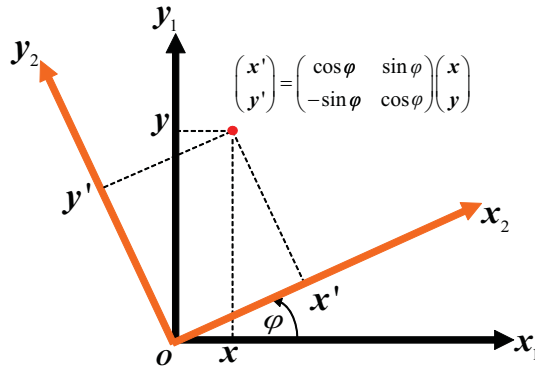


Figure 3.8. coordinate transformation

The KS equation for the case where the two beams form an arbitrary angle ϕ in

the (x, y) plane is given:

$$\begin{aligned}
\frac{\partial h}{\partial t} = & -v_0 + \gamma(1 + j\underline{c}) \left(\frac{\partial h}{\partial x} \right) + \gamma j \underline{s} \left(\frac{\partial h}{\partial y} \right) \\
& + (\xi_x(1 + j\underline{c}^3) + \xi_y j \underline{c} \underline{s}^2) \left(\frac{\partial h}{\partial x} \right) \left(\frac{\partial^2 h}{\partial x^2} \right) + (\xi_x j \underline{c} \underline{s}^2 + \xi_y(1 + j\underline{c}^3)) \left(\frac{\partial h}{\partial x} \right) \left(\frac{\partial^2 h}{\partial y^2} \right) \\
& + 2j \underline{c}^2 \underline{s} (\xi_x - \xi_y) \left(\frac{\partial h}{\partial x} \right) \left(\frac{\partial^2 h}{\partial x \partial y} \right) + 2j \underline{c} \underline{s}^2 (\xi_x - \xi_y) \left(\frac{\partial h}{\partial y} \right) \left(\frac{\partial^2 h}{\partial x \partial y} \right) \\
& + j \underline{s} (\xi_x \underline{c}^2 + \xi_y \underline{s}^2) \left(\frac{\partial h}{\partial y} \right) \left(\frac{\partial^2 h}{\partial x^2} \right) + j \underline{s} (\xi_x \underline{s}^2 + \xi_y \underline{c}^2) \left(\frac{\partial h}{\partial y} \right) \left(\frac{\partial^2 h}{\partial y^2} \right) \\
& + (\nu_x(1 + j\underline{c}^2) + \nu_y j \underline{s}^2) \left(\frac{\partial^2 h}{\partial x^2} \right) + (\nu_x j \underline{s}^2 + \nu_y(1 + j\underline{c}^2)) \left(\frac{\partial^2 h}{\partial y^2} \right) \\
& + 2j \underline{c} \underline{s} (\nu_x - \nu_y) \left(\frac{\partial^2 h}{\partial x \partial y} \right) \\
& + (\Omega_1(1 + j\underline{c}^3) + \Omega_2 j \underline{c} \underline{s}^2) \left(\frac{\partial^3 h}{\partial x^3} \right) + (3\Omega_1 j \underline{c} \underline{s}^2 + \Omega_2(1 + j\underline{c}^3 - 2j \underline{c} \underline{s}^2)) \left(\frac{\partial^3 h}{\partial x \partial y^2} \right) \\
& + j \underline{s} (3\Omega_1 \underline{c}^2 + \Omega_2(\underline{s}^2 - 2\underline{c}^2)) \left(\frac{\partial^3 h}{\partial x^2 \partial y} \right) + j \underline{s} (\Omega_1 \underline{s}^2 + \Omega_2 \underline{c}^2) \left(\frac{\partial^3 h}{\partial y^3} \right) \\
& - (D_{xy}(1 + j\underline{c}^4 - 4j \underline{c}^2 \underline{s}^2 + j \underline{s}^4) + 6D_{xx} j \underline{c}^2 \underline{s}^2 + 6D_{yy} j \underline{c}^2 \underline{s}^2) \left(\frac{\partial^4 h}{\partial x^2 \partial y^2} \right) \\
& - (D_{xx}(1 + j\underline{c}^4) + D_{xy} j \underline{c}^2 \underline{s}^2 + D_{yy} j \underline{s}^4) \left(\frac{\partial^4 h}{\partial x^4} \right) \\
& - (D_{yy}(1 + j\underline{c}^4) + D_{xy} j \underline{c}^2 \underline{s}^2 + D_{xx} j \underline{s}^4) \left(\frac{\partial^4 h}{\partial y^4} \right) \\
& - 2j \underline{c} \underline{s} (2D_{xx} \underline{c}^2 + D_{xy}(\underline{s}^2 - \underline{c}^2) - 2D_{yy} \underline{s}^2) \left(\frac{\partial^4 h}{\partial x^3 \partial y} \right) \\
& - 2j \underline{c} \underline{s} (2D_{xx} \underline{s}^2 + D_{xy}(\underline{c}^2 - \underline{s}^2) - 2D_{yy} \underline{c}^2) \left(\frac{\partial^4 h}{\partial x \partial y^3} \right) \\
& - K \nabla^4 h \\
& + \frac{(\lambda_x(1 + j\underline{c}^2) + \lambda_y j \underline{s}^2)}{2} \left(\frac{\partial h}{\partial x} \right)^2 + \frac{(\lambda_y(1 + j\underline{c}^2) + \lambda_x j \underline{s}^2)}{2} \left(\frac{\partial h}{\partial y} \right)^2 \\
& + j \underline{c} \underline{s} (\lambda_x - \lambda_y) \left(\frac{\partial h}{\partial x} \right) \left(\frac{\partial h}{\partial y} \right) \\
& + \eta(x, y, t),
\end{aligned} \tag{3.11}$$

where $\underline{c} \equiv \cos \phi$, $\underline{s} \equiv \sin \phi$, ϕ is the angle between two incident ion beams in the x - y plane, and $j = J_2/J_1$ is the flux ratio of two beams. All the coefficients in Eq. (3.11) are followed by Makeev's derivation [20].

For the case of $\phi = 90^\circ$, cross derivative terms such as $\partial_x \partial_y h$, $\partial_x^3 \partial_y h$, and $\partial_y^3 \partial_x h$ and the nonlinear term $(\partial_x h)(\partial_y h)$ vanish due to $\underline{cs} = 0$ (cf. Vogel and Linz's approach [46]). Then the equation gives

$$\begin{aligned}
\frac{\partial h}{\partial t} = & -v_0 + \gamma \frac{\partial h}{\partial x} + \gamma j \frac{\partial h}{\partial y} \\
& + \xi_x \left(\frac{\partial h}{\partial x} \right) \left(\frac{\partial^2 h}{\partial x^2} \right) + \xi_y \left(\frac{\partial h}{\partial x} \right) \left(\frac{\partial^2 h}{\partial y^2} \right) \\
& + j \xi_y \left(\frac{\partial h}{\partial y} \right) \left(\frac{\partial^2 h}{\partial x^2} \right) + j \xi_x \left(\frac{\partial h}{\partial y} \right) \left(\frac{\partial^2 h}{\partial y^2} \right) \\
& + (\nu_x + j\nu_y) \frac{\partial^2 h}{\partial x^2} + (j\nu_x + \nu_y) \frac{\partial^2 h}{\partial y^2} \\
& + \Omega_1 \frac{\partial^3 h}{\partial x^3} + \Omega_2 \frac{\partial^3 h}{\partial x \partial y^2} + j\Omega_2 \frac{\partial^3 h}{\partial x^2 \partial y} + j\Omega_1 \frac{\partial^3 h}{\partial y^3} \\
& - (D_{xy}(1+j)) \frac{\partial^4 h}{\partial x^2 \partial y^2} - (D_{xx} + jD_{yy}) \frac{\partial^4 h}{\partial x^4} - (D_{yy} + jD_{xx}) \frac{\partial^4 h}{\partial y^4} \\
& - K \nabla^4 h \\
& + \frac{(\lambda_x + j\lambda_y)}{2} \left(\frac{\partial h}{\partial x} \right)^2 + \frac{(\lambda_y + j\lambda_x)}{2} \left(\frac{\partial h}{\partial y} \right)^2 + \eta(x, y, t). \tag{3.12}
\end{aligned}$$

Also, the coefficients ν , D , and λ are simply transformed as $\nu_x \rightarrow \nu_x^{(\text{DIBS})} = \nu_x^{(x)} + j\nu_x^{(y)}$ and $\nu_y \rightarrow \nu_y^{(\text{DIBS})} = \nu_y^{(x)} + j\nu_y^{(y)}$, where the superscripts (x) and (y) denote the cases where a single ion-beam is incident along the x and the y direction, respectively. D and λ are calculated in the same manner. The numerical value of each coefficient can be estimated by TRIM analysis [70]. For instance, we obtain $a = 23 \text{ \AA}$, $\sigma = 18 \text{ \AA}$, and $\mu = 13 \text{ \AA}$ under ion sputtering of Au(001) substrate with grazing incidence angle of 73° and ion energy $\epsilon = 2 \text{ keV}$. Then, the coefficients of the KS equation are obtained according to Makeev's derivation [20], and these are summarized in Table 3.1. The coefficients for SIBS show strong anisotropy along

the respective ion-beam direction; in the case where the ion beam is injected along the x axis, $\nu_x \gg -\nu_y > 0$ and $-D_{xx} \gg D_{yy} > 0$. In the case of DIBS, such excessive values change the signs of each coefficient: $\nu_x^{(\text{DIBS})}$ and $\nu_y^{(\text{DIBS})}$ become positive, due to the relatively large values of $\nu_x^{(x)}$ and $\nu_y^{(y)}$, and similarly $D_{xx}^{(\text{DIBS})}$ and $D_{yy}^{(\text{DIBS})}$ become negative, due to the large negative values of $D_{xx}^{(x)}$ and $D_{yy}^{(y)}$. In this case, the linear instability for the pattern formation disappears. Our approach, a linear superposition of the respective KS equations corresponding to each independent beam, does not illustrate the interference pattern or the instability shown as the SIBS case. This suggests that a new type of continuum equation is required for explaining the pattern formation during DIBS.

The continuum equation thus obtained remains in the same form as the KS equation if the azimuthal angle between the two incident beams is 90° , although the coefficient of each term is modified to reflect the dual ion beam sputtering. The KS equation reads as follows:

$$\begin{aligned} \frac{\partial h}{\partial t} = & \nu_x \nabla_x^2 h + \nu_y \nabla_y^2 h - D_{xx} \nabla_x^4 h - D_{xy} \nabla_x^2 \nabla_y^2 h - D_{yy} \nabla_y^4 h - K \nabla^4 h \\ & + \lambda_x (\nabla_x h)^2 + \lambda_y (\nabla_y h)^2 + \eta(x, y, t). \end{aligned} \quad (3.13)$$

Here ν is the effective surface tension generated by the erosion process, D is the ion-induced effective diffusion constant [33]. λ is tilt-dependent erosion rate [69], and η is uncorrelated white noise with zero mean, mimicking the randomness resulting from the stochastic nature of ion arrival at the surface [26]. For the KS equation for DIBS, the coefficients ν and D are determined as follows; $\nu_x^{(\text{DIBS})} = \nu_x^{(x)} + j\nu_x^{(y)}$, where $\nu_i^{(x)}$ is the coefficient of the $\nabla_i^2 h$ term when a single ion beam is incident along the x direction, and $j = J_2/J_1$. $\nu_y^{(\text{DIBS})}$ and $D^{(\text{DIBS})}$ are also determined in a similar manner.

The numerical value of each coefficient are estimated by TRIM analysis [70] under the experimental condition [61] and summarized in Table 1. Since D_{xx} and D_{yy} are very large in its absolute magnitude due to sputter-induced diffusion, the

coefficients of $\nabla^2 h$, $\nu_x^{(\text{DIBS})}$ and $\nu_y^{(\text{DIBS})}$, become positive, and $D_{xx}^{(\text{DIBS})}$ and $D_{yy}^{(\text{DIBS})}$ do negative. Thus, the linear instability condition for the KS equation no longer holds for DIBS, in contrast to the experimental observation. Hence, the KS equation for DIBS fails to explain the pattern formation by DIBS.

The contradiction between the stable surface predicted by KS equation for DIBS and the observed nanopattern may be resolved within KS picture if the cross term in the shape of $\partial_x \partial_y h$ exists, because it can yield a different dispersion relation in the linear instability analysis. In fact, the cross term was proposed in Ref. [46] to take into account of the interference effect between the two beams, even though they considered it in the damped KS equation. In our case, we have described DIBS just by adding Sigmund energy dissipation functions of the two beams, and neglected higher order correlations of the sputter events. It should be carefully examined what kind of correlation effects are possible, and how they can be represented in the continuum theory. The cross term can also be obtained in the KS equation for DIBS if the azimuthal angle between the two ion beams are different from 90° . A possibility is that there is usually dispersion in the direction of each ion beam in the experiment, and the angle between them should range some angles around 90° . Then, the cross term may remain effective, and the instability of the surface growth may still hold in DIBS to give the pattern we observed in the experiment.

From theoretical side, we are still cautioned on too simplified treatment of sputtering events during DIBS. During DIBS, each ion beam hits the crossing ripples with large difference in the incidence angles between the illuminated side and shadowed side. For the morphological evolution of such highly modulated surface, a nonlinear term taking into account the dependence of erosion rate, V_o on the local slope was found essential [72]. This suggests that the linear instability argument may not be applicable for the present experimental situation. Finally, redeposition of the sputtered atoms are not included in the KS equation for DIBS. However, recent experimental results [73] and hydrodynamic model [39] and molecular dy-

dynamic simulations [50] predicts their significant contribution to the morphological evolution during ion beam sputtering.

Table 3.1. The coefficients of the KS equation, estimated by the TRIM algorithm under the experimental condition. The presented values are per unit flux.

	SIBS in x (denoted by $^{(x)}$)	SIBS in y (denoted by $^{(y)}$)	DIBS ($j = 1, \phi = 90^\circ$)
ν_x (Å)	14.2	-0.9	$\nu_x^{(x)} + \nu_x^{(y)} = 13.3$
ν_y (Å)	-0.9	14.2	$\nu_y^{(x)} + \nu_y^{(y)} = 13.3$
D_{xx} (Å ³)	-2888.4	38.5	$D_{xx}^{(x)} + D_{xx}^{(y)} = -2849.9$
D_{yy} (Å ³)	38.5	-2888.4	$D_{yy}^{(x)} + D_{yy}^{(y)} = -2849.9$
D_{xy} (Å ³)	-939.1	-939.1	$2D_{xy} = -1878.2$
λ_x (Å ²)	1.40	-0.04	$(\lambda_x^{(x)} + \lambda_x^{(y)})/2 = 0.68$
λ_y (Å ²)	-0.04	1.40	$(\lambda_y^{(x)} + \lambda_y^{(y)})/2 = 0.68$

Chapter 4

Morphological transition on rippled surface by crossing ion-beam sputtering

In the previous chapter, we find that under DIBS each ion beam makes its own pattern so that they interfere one another by disturbing pattern growth by counter beam. The Pattern formation and deformation (smoothing) dynamics coexist in the dual or multiple ion-beam sputtering. Thus, we realized that sputter-induced deformation could be a important issue to go further steps toward understanding full dynamics during ion sputtering.¹

4.1 Introduction

Simultaneous incidence of multiple ion-beams on a surface is limited in its variety by the port arrangement of the sample preparation chamber. Moreover, the precise control of each ion beam to dose the required flux with well-defined incidence angle is not easy, although such control is sometimes demanded as noticed by the high sensitivity of DIBS pattern to the balance of two ion beams (Fig. 4.4).

¹All the experimental works are carried out by Ji-Hyun Kim in Sook-Myung women's university under Prof. Jae-Sung Kim. I contribute this work for data analysis, MD simulation, numerical calculation, and discussion part.

As an alternative, proposed is the sequential application of single ion beam on a sample surface whose orientation with respect to the ion beam is varied in a controlled manner [46]. Vogel and Linz predict through the numerical study of damped Kuramoto-Sivashinsky equation that various nanopatterns can be obtained by sequential ion beam sputtering (SIBS). From the experimental aspects, the orientation of a sample can be quite precisely and reproducibly controlled so that SIBS can be performed in a well-defined manner. For one case of SIBS, we perform crossing ion beam sputtering (CIBS) on the pre-rippled surface (Fig. 4.1).

Here we study how the original structure (say, ripple along y-axis) is decayed, and then new structure (ripple along x-axis parallel to ion beam) is developed under sputtering by crossing (implemented by rotating sample perpendicularly without changing beam geometry) ion beam: (a) What is the major kinetics on deformation/smoothing process by ion sputtering? To find relevant one, scaling behavior of roughness and wavelength in our experimental result is investigated and compared with various models/theories and experiments. (b) The dependence of kinetics on original structure is also investigated with morphology evolution on initially flat surface. (c) To check how well the established Kuramoto-Sivashinsky equation describes particularly on the deformation process. (d) Effect of redeposition; The erosion-based continuum equation for height evolution does not take into account the redeposition effect at all. It's definitely unrealistic when it comes to highly corrugated surface, where sputtered atoms could have more chance to deposit by local curvature than flat surface. To check this effect, we carry out large scale molecular dynamics (MD).

4.2 Experimental

We summarize experimental conditions under CIBS here.

- **Base pressure:** 5×10^{-10} Torr.

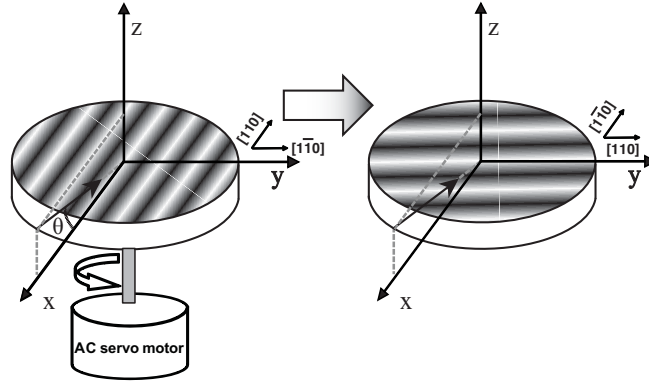


Figure 4.1. Schematic view of sputter geometry and rotatable substrate by the AC servo-motor.

- **Sample:** disk-shaped single crystalline Au(001) with a diameter of 1 cm.
- **Incidence angle:** $\theta = 72^\circ$ from the surface normal. The azimuthal angle $\varphi = [110]$ direction for making initial ripple, then changed to $[1\bar{1}0]$ by rotating sample through AC servomotor (Fig. 4.1).
- **Ion beam flux and energy:** The ion energy $\epsilon = 2.0$ keV (0.5 keV), and flux $f = 0.31$ ions nm^{-2}s (0.06 ions nm^{-2}s) for erosive (diffusive) regime.
- **Duration of sputtering:** It is limited to 5 min. to avoid increasing temperature by the ion sputtering. The sample is maintained at room temperature within maximally 5 K increase.

We observed following quantities to study evolution of surface morphology.

- λ_I : the mean wavelength of the initial ripple. It is obtained well by height-height correlation function in the erosive regime, however in the diffusive regime it is not clear so that we measure it from line profiles.
- λ_N : the mean wavelength of the newly developed ripple.
- A_N : the mean amplitude of the newly developed ripple, obtained from numerous line profiles.

4.3 Morphological features in CIBS

Here we summarize morphology evolution in CIBS.

4.3.1 CIBS in the erosive regime

- Fig. 4.2(a): Initial ripple pattern obtained by ion beam sputtering.
- Fig. 4.2(b): Initial ripple seriously destroyed by CIBS. Then emerged is new ripple oriented along with the crossing ion beam.
- Fig. 4.2(b-1): The newly developed ripple preferred the illuminated side.
- Fig. 4.2(b-2): No discernable ripple appeared on the shadow side. The reason seems to be larger sputter yield in the illuminated side, plus larger flat area than the shadow side so that the new ripple grows well without disturbing.
- Fig. 4.2(c): Initial ripple almost smeared out ($A_I < 5 \text{ \AA}$ and $\lambda_I > 450 \text{ nm}$). While, new ripples grown continually, as a result $A_N = 6.1 \text{ \AA}$ and $\lambda_N = 22 \text{ nm}$. The coherence length of the new ripple was longer than 450 nm, and was limited merely by the remnants of the initial ripple.
- Fig. 4.2(d): Only the new ripple is covered on the surface, which have a coherence length longer than $3 \mu\text{m}$. Stationary state is likely to be reached.

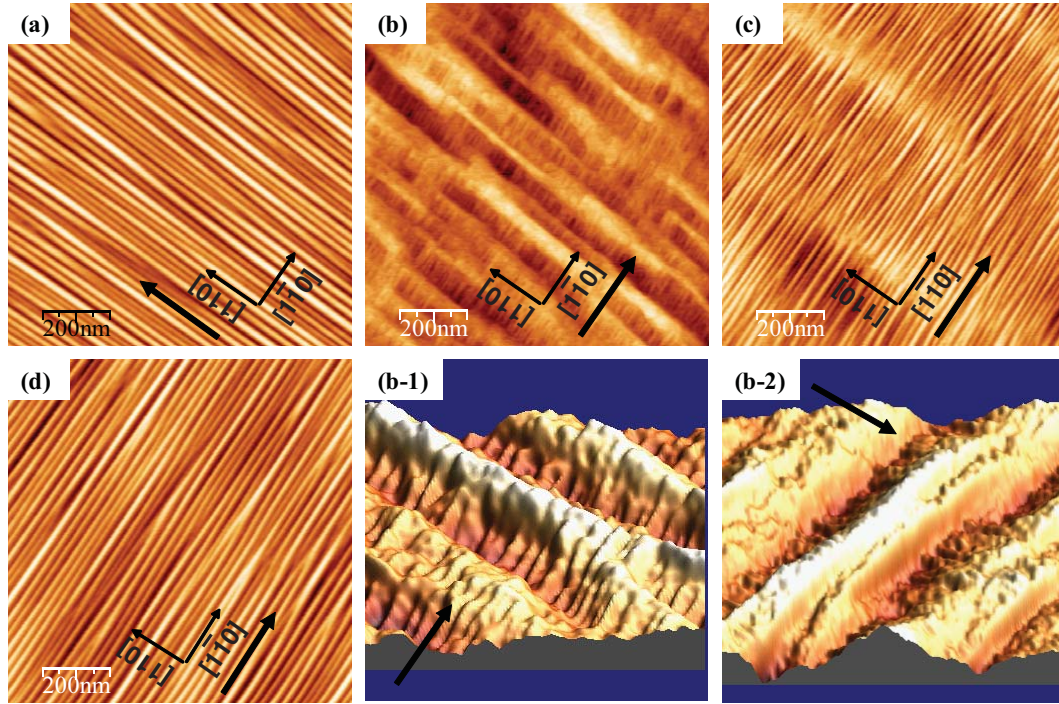


Figure 4.2. (color) AFM images ($1 \times 1 \mu\text{m}^2$) of the surface after sputtering in the erosive regime. (a) Initial rippled surface generated by ion beam sputtering. Surface morphology after CIBS at (b) $\Psi = 75.0 \text{ ions nm}^{-2}$, (c) $\Psi = 468.8 \text{ ions nm}^{-2}$, and (d) $\Psi = 1875.0 \text{ ions nm}^{-2}$. (b-1) and (b-2) are three-dimensional, detailed images ($457 \times 457 \text{ nm}^2$) of a representative area in (b); (b-1) illuminated and (b-2) shadowed side of the ripples. All the arrows within the figures indicate the ion beam direction.

4.3.2 CIBS in the diffusive regime

- Fig. 4.3(a): Initial ripple pattern obtained by ion beam sputtering, which have larger $\lambda_I = 20\text{--}50\text{ nm}$ and $A_I = 5\text{--}41\text{ \AA}$ than those in the erosive regime due to coalesce frequently between neighbor ripples by effective surface diffusion.
- Fig. 4.3(b): Surface morphology at $\Psi = 153.8\text{ ions nm}^{-2}$. W decreased to $11.3\text{ \AA}$ and λ_I increased to 52 nm .
- Fig. 4.3(b-1): a 3D view of bridging structure, which connect adjacent initial ripples in Fig. 4.3(b). The distribution of distance of bridges is wide as $50\text{--}130\text{ nm}$, which is very large in comparison with $\lambda_N = 27\text{ nm}$ in [Fig. 4.3(d)]. Combined initial ripples with the bridging structures, a rectangular patterns of various sizes reveal, which is very unique characteristic of the diffusive regime.
- Fig. 4.3(c): $\Psi = 716.3\text{ ions nm}^{-2}$. W decreased to 6.8 nm , and $\lambda_I \approx 150\text{ nm}$ increased remarkably.
- Fig. 4.3(c-1): At the same time, the rectangular pattern enlarged, and therein, new ripples nested densely with λ_N varying from $20\text{ to }40\text{ nm}$ and amplitude A_N varying from $2\text{ to }10\text{ \AA}$
- Fig. 4.3(d): Finally, prolonged sputtering resulted in the disappearance of the initial ripple, and the new ripples prevailed over the surface ,characterized by $\lambda_N \approx 27\text{ nm}$ and $W \approx 8\text{ \AA}$. λ_N and W did not change.

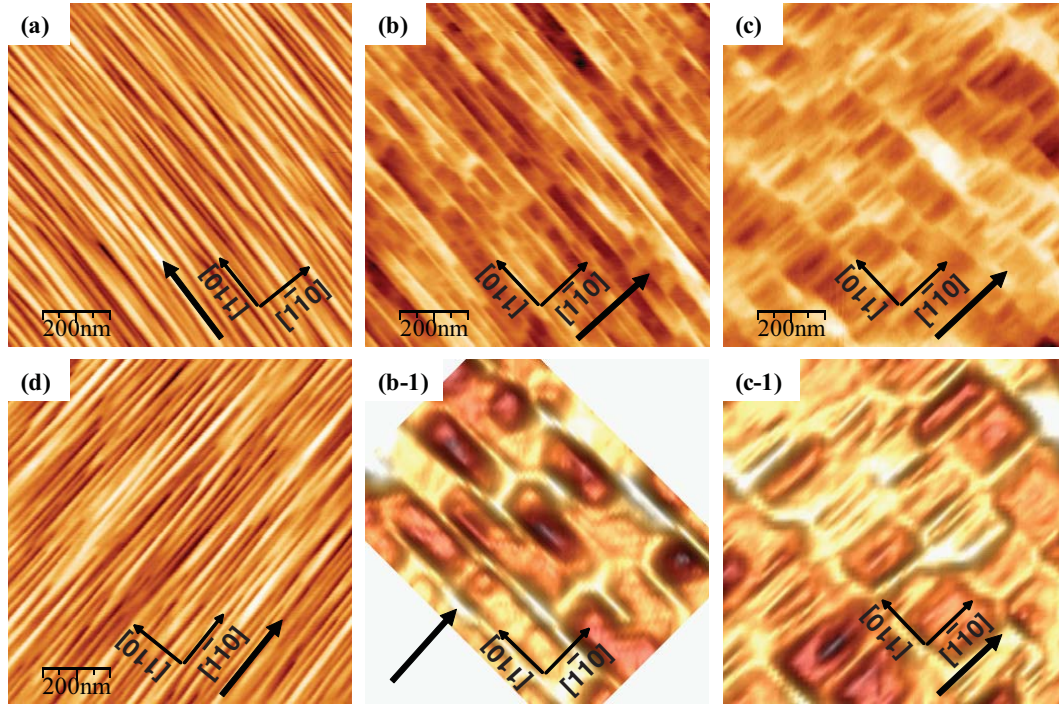


Figure 4.3. (color) AFM topographies ($1 \times 1 \mu\text{m}^2$) after sputtering in the diffusive regime. (a) Initial rippled surface. Surface morphology after CIBS at (b) $\Psi = 153.8 \text{ ions nm}^{-2}$, (c) $\Psi = 716.3 \text{ ions nm}^{-2}$, and (d) $\Psi = 7128.8 \text{ ions nm}^{-2}$. (b-1) and (c-1) are three-dimensional detailed images ($318 \times 238 \text{ nm}^2$, $560 \times 540 \text{ nm}^2$); (b-1) is a representative rectangular pattern in (b), and (c-1) is a representative new ripple nested in a rectangular pattern in (c). All of the arrows in the figures indicate the ion beam direction.

4.4 Scailing behavior in CIBS

Figs. 4.4 and 4.5 indicate how surface roughness W and ripple wavelength λ evolve in time under CIBS in the erosive and diffusive regime, respectively. The scailing behaviors are summarized in table 4.1. We refer some unique features in CIBS.

- **High growth rate in W or A_N :** In the erosive regime, the growth exponent β is 0.96. Exponential fitting also well reproduce the data. The roughness involves both effect of new ripple growth and decay of the initial ripple. Thus, in order to see sole effect of new ripple, we introduce mean amplitude of the new ripple A_N in which contribution of the initial ripple is subtracted from each line profile. Then A_N grows as power law with $\beta \sim 1.38$, larger than that of W [Fig. 4.4(a)]. This value of β is very large compared to $\beta \sim 0.38$ for the initially flat surface under the same sputtering conditions [Fig. 4.6(a)], indicating that the growth kinetics of the new ripples on the pre-rippled surface is quite different from that on the initially flat surface. Moreover, no model predicts the high $\beta \geq 1$ observed during the growth of the new ripple [24].
- **Correlation between λ_N growth and new ripple development:** λ_I is almost constant in the early stages of CIBS. However, λ_I abruptly increases between $\psi = 37.5$ ions nm^{-2} and $\psi = 65.6$ ions nm^{-2} [the dotted region in Fig. 4.4(b)], and then follows a power law with a coarsening exponent $1/z = 0.63 \pm 0.09$. The start of the transition period coincides with the start of the new ripple development. New ripple enhance surface diffusion along the ripple's step edge, which could speed up the coarsening process.
- **λ_N behavior:** In Fig. 4.4(c). The dashed blue line represents a power-law fit $\lambda_N \sim t^n$ with coarsening exponent $n = 1/z = 0.13 \pm 0.01$, small in comparison with the values predicted by various models [24]. Moreover, we can well fit the experimental results with a logarithmic function (solid red

line). Such coarsening is incompatible with the prediction of the BH and Kuramoto-Sivashinsky equations.

- W and λ_I in diffusive regime:** In Fig. 4.5(a), W decays exponentially in the early stages of CIBS, similar with the erosive regime. However as new ripples develop at increasing fluence, W starts to increase. In the diffusive regime W and A_N seems to reach a stationary state within at most a decade, we neglect to discern them. In Fig. 4.5(b), λ_I is almost constant until $\psi = 60.0 \text{ ions nm}^{-2}$, but it then starts to increase (the red dotted region). After a transition period, λ_I increases, following a power law behavior with $1/z = 0.74$, as was the case for the erosive regime in Fig. 4.4(b). Compared with the erosive regime, the transition period is longer and λ_I increase during that period is gentle. In detail view of the transition period, there is slow increase in early stage then later rapid growth. The branching points are related with the development of a bridging structure ($\psi = 60.0 \text{ ions nm}^{-2}$) and of a new ripple ($\psi = 153.8 \text{ ions nm}^{-2}$), respectively. In the diffusive regime, new ripples coexist with the bridging structures and frequently merge with neighboring ones. Thus, λ_N is not well-defined.
- Comparison with sputtering on flat surface:** Under same sputtering condition as those in the erosive regime, but initially flat surface is used. In figure 4.6(a), W_F (subscript F denotes flat surface) remains almost constant or fluctuating until $\psi = 281.3 \text{ ions nm}^{-2}$. After the incipient stage, W_F increases, closely following the power law with $\beta = 0.38 \pm 0.04$. It is much smaller than that observed during CIBS . In Fig. 4.6(b), coarsening of ripple wavelength λ_F shows a crossover from $1/z = 0.3$ to 0.06. This show very distinctive feature compared with the results in the erosive regime, implying that the growth kinetics of ripples on the initially flat surface is different from that on the pre-rippled surface.

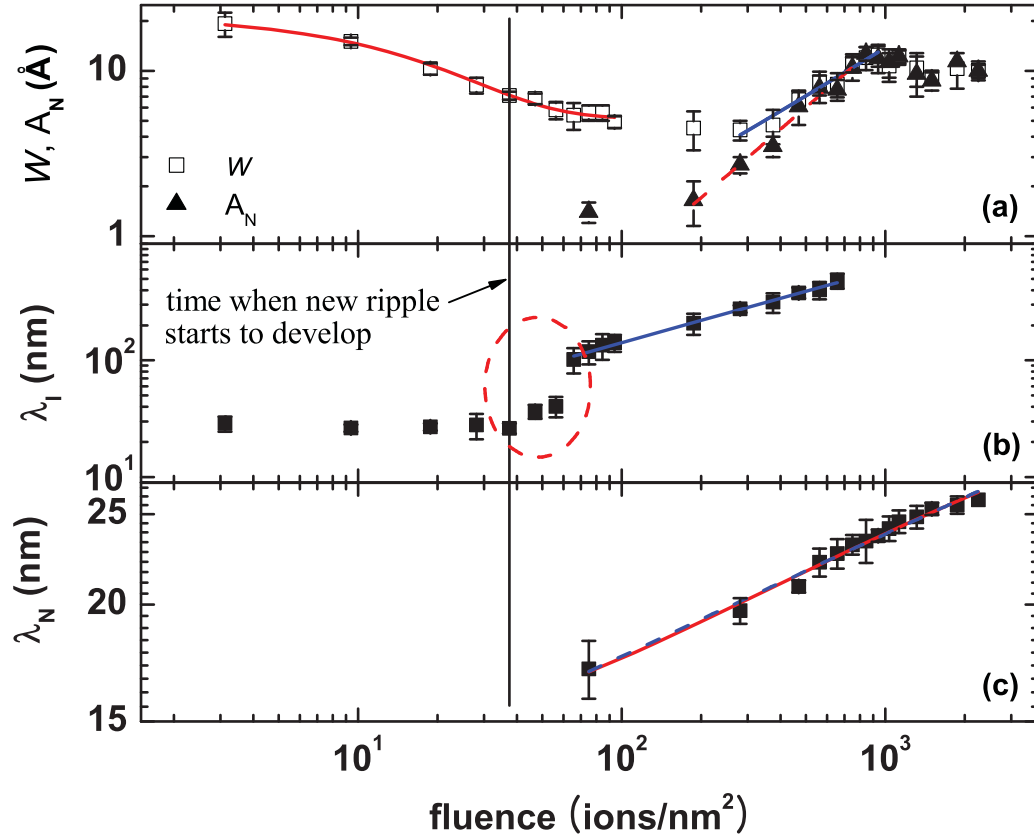


Figure 4.4. (color) Evolution of surface morphology by ion beam sputtering in the erosive regime. (a) Surface roughness W and new ripple amplitude A_N vs. ion fluence. The solid red line (on left-hand side) represents an exponential fit to W during its decay, and the blue line (on the right-hand side) represents a power law fit to W during its growth, with growth exponent $\beta = 0.96 \pm 0.09$. The dashed red line corresponds to a power law fit to A_N with $\beta \sim 1.38 \pm 0.12$. (b) Wavelength of the initial ripple λ_I vs. ion fluence. The solid blue line denotes a power law fit with coarsening exponent $1/z = 0.63 \pm 0.09$. (c) Wavelength of the new ripple λ_N vs. ion fluence. The solid red line denotes a logarithmic fit, and the dashed blue line denotes a power law fit with coarsening exponent $1/z = 0.13 \pm 0.01$.

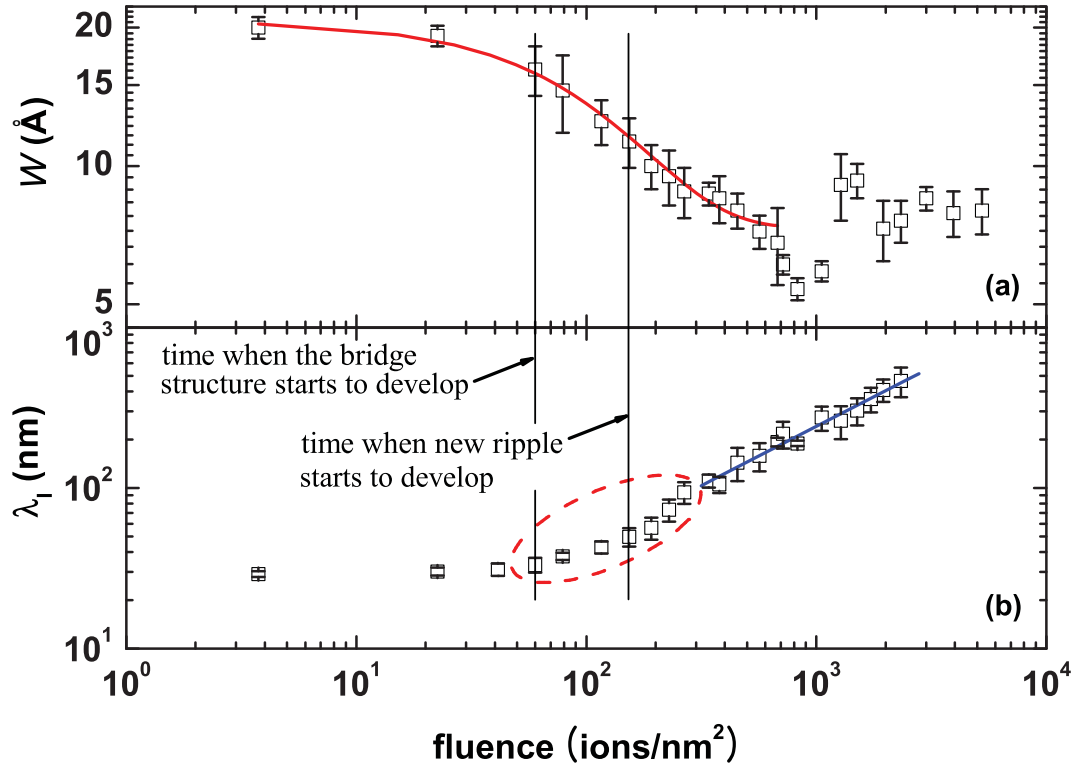


Figure 4.5. (color) Evolution of surface morphology by ion sputtering in the diffusive regime. (a) Roughness W vs. ion fluence. The solid line in the decay portion denotes an exponential fit to W . (b) Wavelength of the initial ripple λ_I vs. ion fluence. The solid blue line denotes a power-law fit to λ_I with a coarsening exponent $1/z = 0.74 \pm 0.04$.

4.5 Discussion

4.5.1 Sputter-induced smoothing

Exponential decay of W was observed in the early stages of CIBS in both the erosive and diffusive regimes, as shown in Figs. 4.4(a) and 5(a), respectively. An inverse linear fit gave rise to χ^2 values that were at least six times greater than those obtained from the exponential fit in both regime. Similar sputter-induced smoothing was also found in previous studies. Fan *et al.*, [93] studied the morphological evolution of Si nanodots by ion beam sputtering, and also observed the exponential decay of the dot heights. The numerical integration of the KS equation was used to elucidate the observed morphological evolution, but did not yield satisfying results. However, the temporal evolution of the nanodot pattern [93] and its dependence on the incident ion flux [94] could be better described by adding a damping term, $-\alpha h$, to the KS equation, known as the damped KS equation (dKS) [95]. The damping term was partly attributed to the redeposition of sputtered atoms.

In line with the above argument, Toyoda *et al.* observed that the initially rippled surface underwent exponential decay of its roughness by cluster ion beam sputtering [51]. They introduced a model in which some of the sputtered atoms redeposited, diffused toward the bottom of the valleys, and then filled them. This picture was supported by their kinetic Monte Carlo (MC) simulations. The research of both Fan *et al.* and Toyoda *et al.* suggested that redeposition of sputtered atoms and their diffusion into the valleys plays a significant role in the sputter-induced smoothing of the present pre-rippled surface.

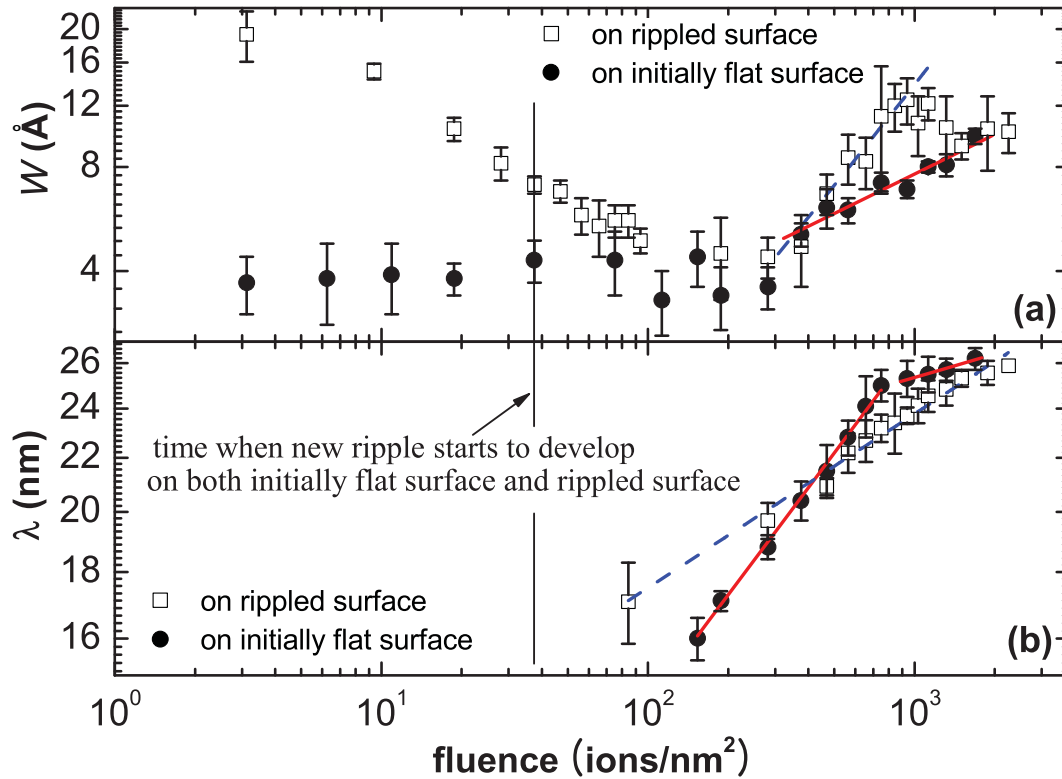


Figure 4.6. (color) Comparison of the morphological evolution from the initially flat surface (filled circle) when ion sputtering is performed in the erosive regime, with that from the pre-patterned surface (empty square). (a) Roughness W vs. ion fluence. The solid red line (dotted blue line) denotes a power-law fit to W of the initially flat surface (the pre-patterned surface, as previously noted in Fig. 4.4(a)) with a growth exponent $\beta = 0.38 \pm 0.04$ ($\beta = 0.96 \pm 0.09$). (b) New ripple wavelength λ vs. ion fluence. The solid red line (dotted blue line) denotes a power-law fit to λ of the initially flat surface (pre-patterned surface, as previously noted in Fig. 4.4(c)) with a coarsening exponent $1/z = 0.30 \pm 0.01$ before the crossover and then $1/z = 0.06 \pm 0.01$ ($1/z = 0.13 \pm 0.01$).

To examine this picture more concretely, we performed a classical molecular dynamics (MD) simulation of CIBS on a pre-rippled surface modeled after a sinusoidal wave with the same mean amplitude and wavelength as those observed in Fig. 4.2(a). We used the embedded atom method (EAM) potential for Au [96] and the Ziegler, Biersack and Littmark (ZBL) potential [97] to describe the highly repulsive behavior at a short distance. Utilizing the switching function, the pairwise potential in EAM was modified to smoothly join the ZBL potential within 2 Å of the atomic distance, thus avoiding splining errors and the derivative discontinuity problem [98,99]. With this method, the sputter erosion and redeposition were rigorously treated, although the thermal diffusion of redeposited adatoms was not fully taken into account.

Figure 4.7(a) shows a map of the height change after sputtering the initial sinusoidal ripple. Although the total fluence was limited to $\Psi = 0.96$ ions nm⁻² due to computational limitations, we clearly observed the morphology variation. Along with the initial ripples, Fig. 4.7(b) shows the height profiles of the sputtered ripples after averaging along each ripple. The height of the sputtered ripple on the illuminated side and near its top decreased due to sputter erosion. Note that the redeposition substantially contributed to the morphological evolution. The height of the shadowed side significantly increased due to the redeposition. Some of the sputtered atoms also landed on the valley and slightly increased the valley height. Overall, the mean amplitude of the ripple decreased, and the rippled surface eventually was smoothed by sputter erosion and redeposition. If the diffusion of the redeposited atoms were taken into account, smoothing of the rippled surface would be further expedited. These results supported Toyoda's model [51] and the observations of Fan *et al.* [93] that indicated the significance of redeposition but contradicted the BH theory prediction of sputter erosion causing the instability [21].

Erlebacher *et al.* studied the smoothing of rippled Si(001) through thermal annealing [105]. In their work, the amplitude did not follow the classical theory that

predicted an exponential decay with time, [22, 23] but did follow an inverse linear decay. This behavior was in accordance with what Rettori and Villain [100] and Ozdemir and Zangwill [101] predicted. In those models, the morphological evolution was mainly governed by step-traversal limited kinetics. Sputter-induced morphological changes shared many aspects in common with the thermal smoothing sample because sputtering generated adatoms as well as vacancies, and their diffusion, contributed to the shaping of the surface. Moreover, our surface was a crystalline surface such that the step crossing over the Ehrlich-Schwoebel (ES) barrier by both adatoms and vacancies played an important role in the morphological evolution. Figure 4.4(a), however, showed that the inverse linear function had a relatively poor fit compared with an exponential fit. This indicated that processes other than thermal diffusion, limited by ES barriers such as redeposition and slope-dependent erosion [as shown in Fig. 4.7(b)], play major roles during the sputter-induced smoothing.

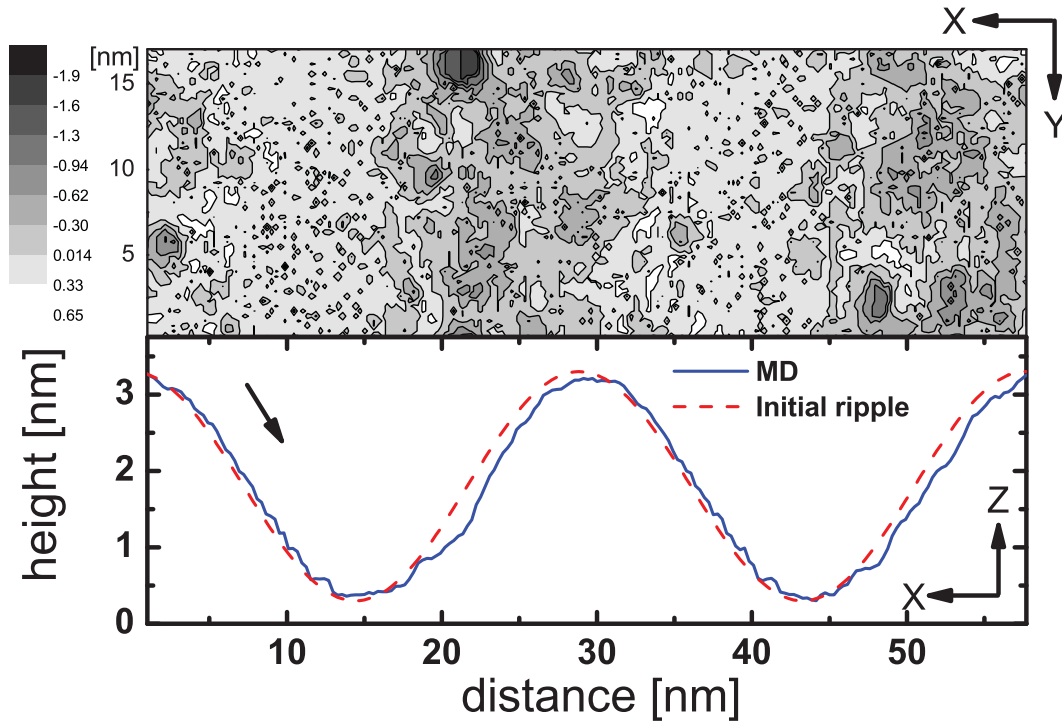


Figure 4.7. (color online) Surface profiles after $0.96 \text{ ions nm}^{-2}$ of Ar^+ sputtered across the ripples with an incidence energy of 2.0 keV. (a) Plot of the height change after sputtering. Negative values denote erosion, while positive values denote an increase of the surface height due to redeposition. (b) Profiles of the rippled surface before and after sputtering. Profiles of the ripples after sputtering are obtained from (a) after averaging along the direction of each ripple or perpendicular to the train of ripples. The arrow in (b) denotes the direction of the incident ion beam.

4.5.2 Coarsening of the initial ripple

Figures 4.4(b) and 4.5(b) show the evolution of λ_I in both the erosive and diffusive regimes, respectively. λ_I was constant in the early sputtering stages in both regimes. In the erosive regime, λ_I showed an abrupt increase when the new ripples developed. [See the transition region indicated by the dotted red ellipse in Fig. 4.4(b).] These results suggest that the new ripples played a significant role in the coarsening of the initial ripple, thus offering very efficient channels mediating mass transport between adjacent initial ripples. The step edges of the new ripples in the close-packed [110] direction could be the channels because the diffusion barrier encountered by adatoms along the edges was very low. As a result, ripening of one ripple occurs at the expense of the other, thus increasing λ_I . Before the new ripples could develop, each initial ripple was isolated, and little coarsening of the initial ripple was observed.

In the diffusive regime, however, the coarsening of the initial ripples started before the new ripples could develop (Fig. 4.5). A key difference from the erosive regime was the formation of structures bridging neighboring ripples before a new ripple developed [Fig. 4.3(b) and (b-1)]. This bridge could play the same role as a new ripple did in mediating mass transport between adjacent initial ripples.

In Fig. 4.5, the coarsening rate during the bridge forming stage was, however, lower than that during the ripple forming stage because the density of the bridge structure was lower than that of the new ripples. For the initial ripple in the later stages, the coarsening exponent $1/z$ was 0.63 and 0.74 for the erosive and the diffusive regimes, respectively. On the other hand, $1/z$ for the thermal coarsening of mounds is usually less than 0.5 [102–104]. The coarsening exponents observed in the present experiments were distinctly higher than those observed during the thermal coarsening of mounds. This suggests that the coarsening of the initial ripples was mediated by very efficient diffusion channels, or by ripple-mediated diffusion along its step edges that is a characteristic of the pre-rippled surface but was not

available in the usual two-dimensional coarsening of the mounds.

4.5.3 Kinetic behaviors on the pre-rippled surface versus those on the initially flat surface

After reaching its minimum, W increased as the new ripples developed (Figs. 4.4 and 4.5). In the erosive regime during CIBS [Fig. 4.4(a)], W followed the power law with a growth exponent $\beta = 0.96 \pm 0.09$. This was much higher than the $\beta = 0.38$ observed for an initially flat Au(001) under the same sputtering conditions for CIBS [Fig. 4.6(a)]. Note that there were some differences between the two cases. On the pre-rippled surface in the early stages, a new ripple developed preferentially on the extended flat area of the illuminated side of the initial ripples [Figs. 4.2(b) and 2(b-1)]. Thus, the local incidence angle of the ion beam, θ [Fig. 4.1(a)], was smaller for the pre-rippled surface than for flat surface. Consequently, the sputter yield was larger on the pre-rippled surface than on the flat surface. This, in turn, gives rise to a morphological evolution or growth exponent for the pre-rippled surface that differs from that for the flat surface.

During CIBS in the erosive regime, λ_N of the new ripples followed a logarithmic behavior, represented by the solid (red) line in [Fig. 4.4(c)], or a power law with a very low coarsening exponent, $1/z = 0.13 \pm 0.01$, represented by a dashed (blue) line in [Fig. 4.4(c)]. On the initially flat surface, λ_F followed a power law with $n \sim 0.3$ in the early stages. In the later stages, however, the surface became as rough as the pre-rippled surface, and showed coarsening behavior with small $n \sim 0.06$, or logarithmic behavior similar to the pre-rippled surface [Fig. 4.6(b)]. The crossover behavior in the ripple coarsening on the initially flat surface seemed to be triggered by the roughening of the initially flat surface by the continued sputtering. Similar crossover behavior was previously observed: as sputtering proceeded, W of an initially flat Pd(001) changed from an initial power law behavior to a logarithmic behavior in the later stages [66]. The crossover behavior was attributed to the en-

hanced probability of redeposition because the capture angle of the sputtered atoms increased for the roughened surface. In this figure, the coarsening behavior of the pre-rippled surface or the logarithmic behavior from the beginning of the sputtering process can be understood in the following manner: the initial surface was rough from the beginning and the redeposition was already significant. For the initially flat surface, logarithmic behavior set in only at the later stages, after the surface became rough and the redeposition became significant.

In an attempt to elucidate the morphological evolution of the pre-rippled surface treated by CIBS, we conducted numerical studies with the KS equation. The input parameters varied within the parameter space suggested by Makeev *et al.* [20]. The ripple pattern was generated with its orientation parallel to the ion beam direction, as observed in our experiment. We also examined the parameters predicted by both an SRIM simulation and our own MD simulation. However, we failed to reproduce the experimentally observed evolution of ripple patterns. Although the importance of the redeposition was indicated in previous discussions, a term $\nabla^2(\nabla h)^2$ that derived from consideration of redeposition [39], missing in the KS equation. Thus, the failure of the numerical simulation might be taken as evidence of the significance of disregarded effect such as the redeposition.

Table 4.1. Summary of scaling behavior in CIBS.

	Erosive		Diffusive		Flat	
	early	later	early	later	early	later
W	exp. decay	$\beta = 0.96$ (exp. growth)	exp. decay	-	$\beta = 0.38$	
A_N	-	$\beta = 1.38$	-	-	-	
λ_I	const.	$1/z = 0.63$	const.	$1/z = 0.74$	-	
λ_N	-	$1/z = 0.13$ (log.)	-	-	$1/z = 0.3$	$1/z = 0.06$ (log.)

Chapter 5

Conclusion

This thesis has focused on understanding the pattern formation by ion-beam sputtering, especially dual ion-beam sputtering. In order to do this, we have studied various theories and models essentially starting from the Sigmund theory in Chapter 1. Also an experimental approach is used. We selected the Au(001) as a target surface because it is known to be stable at the ambient condition, and we can thereby obtain reliable *ex situ* AFM (atomic force microscope) images and study the time evolution or the pattern growth of surface morphology under various sputter conditions.

In Chapter 2, we sputtered Au(001) by a single ion beam at a grazing angle and studied dependence of the pattern formation on several sputter conditions such as ion beam flux, fluence, energy, and azimuthal angle between the high axis of the substrate and ion beam projection. By tuning such sputter conditions, controlling or designing the sputter-induced patterns is investigated. To obtain a well-defined ripple structure, a sufficiently large flux and match of high symmetry axis of the substrate (e.g. [110] axis in (001) surface) to the in-plane ion beam direction are required. The ion energy and fluence are largely involved in the growth of the pattern, but not in the order of the pattern. Under our sputter parameter regimes associated with metal substrates (which maintains its crystallinity under sputtering, contrary to semiconductor e.g. Si), ripple orientation depends on the competition between the ion beam direction (when erosion is dominant) and the crystallographic direction

(when surface diffusion is dominant). At an intermediate regime, the effects of both are clearly visible in terms of the degradation of ripple ordering.

In Chapter 3, we showed for the first time how the simultaneous incidence of a dual ion beam could influence pattern formation on the sputter-eroded surface. To obtain a diverse pattern with ion beam sputtering, it was suggested by Carter [45] and Linz [46] that sputtering with dual or multiple ion sources, having different sputter geometry respectively, could bring a novel interference pattern by controlling ion-beam geometry. To realise these proposals, the first step was achieved experimentally by dual ion-beam sputtering (DIBS) on Au(001).

Highly ordered nanopatterns were formed by DIBS: a two-dimensional square-symmetry nanohole array in the diffusive regime and a nanodot array in the erosive regime. In contrast with the diffusive regime, a surface instability emerged and is triggered by the interplay between two ion beams: detailed balancing of the two crossing ion beams is necessary for the 2-D pattern formation, otherwise 1-D modulated ripples prevailed.

We observed many salient features such as the slope asymmetry of ripple and morphological transition from ripple to dot by controlling each flux. Such phenomena were newly reported, and thus we expect that our study will stimulate succeeding studies on pattern formation by employing multiple ion sources.

A new experiment has a role of being a test bed of an existing theory. We firstly extended a continuum equation for the surface height evolution responsible for the dual ion-beam sputtering, starting from the Kuramoto-Sivashinsky (KS) equation. The derived equation is inherently the same as the KS equation, although the coefficients are adjusted to reflect DIBS. It predicts a stable surface, contrary to the pattern formation as well as the experimental observations.

In the theoretical aspect of DIBS, the Bradley-Harper (BH) approach failed to describe the pattern or surface instability invoked by DIBS. Challenging issues to the conventional continuum models have arisen from the distinctive results of DIBS

such as

- Local slope asymmetry
- Flux-triggered dot-to-ripple transition in the erosive regime

To describe accurately multiple-ion-beam sputtering and pattern formation, it may be necessary to take into account factors that are currently missing from the BH model (e.g., the correlation effects of ion beams and redeposition). At present, Cuerno's hydrodynamic model could be an alternative explanation of the phenomena induced by DIBS because of its enriched dynamics. However, we need to develop it further to take into account a more realistic sputter condition beyond ion characteristics e.g. finding the redeposition rate as a function of sputter condition and local geometry, and construction of a relevant continuum model to represent the effects of a crystalline (metal) surface, such as the Ehrlich-Schwoebel barrier.

These limitations might indicate that dual or multiple ion beam sputtering implies intrinsically different kinetics from those caused by single ion-beam sputtering. For instance, one ion beam in DIBS has duality: making its own pattern and thereby disturbing the pattern growth by another beam. Thus, the pattern formation and deformation/smoothing dynamics coexist in the dual or multiple ion beam sputtering. We discussed this more in the following chapter.

In Chapter 4, we carried out a comprehensive study on the morphological evolution of a pre-rippled surface under crossing-ion-beam sputtering (CIBS). In this situation, the initial ripple gets eroded more and more, and eventually fades away as the CIBS proceeds. At an intermediate time, the ripple deformation could provide valuable information as to what happens on the surface by utilising the interplay between the original ripple structure and newly emerged ripple.

In both the erosive and diffusive regimes, the amplitude of the pre-rippled surface decayed exponentially with time by the erosion of the crossing ion beam; at the same time, new ripples developed parallel with the ion beam direction and appeared

only after the rippled surface became flat. As a result, a superimposed pattern consisting of the initial and new ripple patterns was not observed, contrary to a recent theoretical prediction [46].

In contrast to the prediction of the BH theory, sputtering smoothed the surface for the pre-rippled surface. The smoothing could be proceeded via non-thermal processes, such as redeposition and slope-dependent erosion. With increasing ion fluence of the crossing ion beam, the coarsening of the initial ripple followed a power law. The coarsening exponent $1/z$ was 0.63 in the erosive regime and 0.74 in the diffusive regime. These values were much larger than those observed during the thermal coarsening of two-dimensional mounds, and implied that a very efficient mass transport mechanism was involved in the CIBS. We suggested a ripple-mediated transport mechanism where mobile defects, generated during the sputtering, effectively diffuse via one-dimensional structures bridging adjacent ripples.

New ripples in the erosive regime showed a logarithmic coarsening of the ripple wavelength with time. In contrast with the initially flat surface, ripple coarsening reveals a crossover behaviour from the power law to the logarithmic behaviour although the sputtering conditions are same. The logarithmic behaviour might be attributed to increased redeposition in the latter stages due to the roughened surface. Molecular dynamics (MD) simulation mimicking the CIBS also revealed that redeposition of sputtered atoms on the valleys and shadowed slopes had a significant role in the evolution of surface morphology.

The BH and KS models could not explain the observed morphological evolution of the pre-rippled surface e.g. the coarsening behaviour. Since the hydrodynamic model [39], which takes into account the redeposition effects and predicted the logarithmic coarsening of ripples, this model is expected to help interpret the present experimental results. Such a study is currently in progress.

Appendix A

Numerical recipes

A.1 Reduced form of the KS equation

The KS equation, including conserved KPZ (CKPZ) term, is given by

$$\frac{\partial h}{\partial t} = \nu \nabla^2 h - K \nabla^4 h + \frac{\lambda_1}{2} |\nabla h|^2 + \frac{\lambda_2}{2} \nabla^2 (\nabla h)^2 + \eta, \quad (\text{A.1})$$

where the noise term η is a spatiotemporally uncorrelated white noise satisfying both

$$\langle \eta(\vec{r}, t) \rangle = 0, \quad (\text{A.2})$$

and

$$\langle \eta(\vec{r}, t) \eta(\vec{r}', t') \rangle = 2D \delta(\vec{r} - \vec{r}') \delta(t - t'). \quad (\text{A.3})$$

The $2D$ in Eq. (A.3) is given by a variance of the noise term. If we choose η as a uncorrelated uniform noise $R(\vec{r}, t)$ within $[-0.5, 0.5]$, the variance of R is as follows:

$$\langle R^2(\vec{r}, t) \rangle - \langle R(\vec{r}, t) \rangle^2 = \langle R^2 \rangle = \int_{-0.5}^{0.5} R^2 \cdot (P(R) = 1) dR = \frac{1}{3} R^3 \Big|_{-0.5}^{0.5} = \frac{1}{12} \quad (\text{A.4})$$

For numerical calculation of Eq. (A.1), we can write it by using discretized variables as

$$\frac{\Delta h}{\Delta t} = \nu \nabla^2 h - K \nabla^4 h + \frac{\lambda_1}{2} |\nabla h|^2 + \frac{\lambda_2}{2} \nabla^2 (\nabla h)^2 + \left(\frac{2D \cdot 12}{\Delta t \Delta r^2} \right)^{1/2} R(\vec{r}, t). \quad (\text{A.5})$$

The coefficient of R is chosen so that Eq. (A.3) is satisfied. The denominator $\Delta t \Delta r^2$ is present due to the delta function, the spatial temporal correlation. The $2D$ comes from straight from Eq. (A.3) and the additional factor of 12 is due to the fact that R is a uniform noise distributed between -0.5 and 0.5 , thus has a variance of $1/12$. Furthermore, we can make the KS equation simpler by rescaling it; introducing dimensionless variables as

$$h' = h/h_0, \quad \vec{r}' = \vec{r}/r_0, \quad t' = t/t_0, \quad g_1 = \nu \lambda_2 / K \lambda_1, \quad g_2 = D \lambda_1^2 / \nu^3. \quad (\text{A.6})$$

We can check that g_1 and g_2 are dimensionless because the dimension of ν , K , λ_1 , and D are $[m^2/s]$, $[m^4/s]$, $[m/s]$, and $[m^4/s]$, respectively. The ∇ and its second and fourth order derivative become $(1/r_0)\nabla'$, $(1/r_0^2)\nabla'^2$ and $(1/r_0^4)\nabla'^4$, respectively. Then we can write the Eq. (A.1) in primed dimensionless variables as

$$\begin{aligned} \frac{\Delta h'}{\Delta t'} &= \frac{\nu t_0}{r_0^2} \nabla'^2 h' - \frac{K t_0}{r_0^4} \nabla'^4 h' + \frac{\lambda_1 h_0 t_0}{2 r_0^2} |\nabla' h'|^2 + \frac{\lambda_2 h_0 t_0}{2 r_0^4} \nabla'^2 (\nabla' h')^2 \\ &+ \left(\frac{1}{t_0 r_0^2}\right)^{1/2} \left(\frac{2D \cdot 12}{\Delta t' \Delta r'^2}\right)^{1/2} R(\vec{r}', t'). \end{aligned} \quad (\text{A.7})$$

Without losing generality, we can express the scaling factors by combination of the parameters like

$$h_0 = \nu / \lambda, \quad r_0^2 = \nu t_0 (= -|\nu| t_0 \text{ if } \nu < 0), \quad t_0 = K / \nu^2. \quad (\text{A.8})$$

Then all the coefficients in the Eq. (A.7) can be eliminated except for the coupling constant g_1 and g_2 . Taken unprimed, the equation gives

$$\frac{\Delta h}{\Delta t} = \nabla^2 h - \nabla^4 h + \frac{1}{2} |\nabla h|^2 + \frac{g_1}{2} \nabla^2 (\nabla h)^2 + (g_2)^{1/2} \left(\frac{24}{\Delta t \Delta r^2}\right)^{1/2} R(\vec{r}, t). \quad (\text{A.9})$$

Now it is nothing but the initial value problem which can be solved numerically with given initial surface morphology.

Bibliography

1. M. Navez, C. Sella and D. Chaperot, *Microscopie electronique étude de l'attaque du verre par bombardement ionique* C. R. Acad. Sci., Paris **254**, 240 (1962).
2. For a review, for example, see L. Jacak, P. Hawrylak and A. Wojs, *Quantum Dots* (Springer-Verlag, Berlin, 1998).
3. J. V. Barth, G. Costantini, and K. Kern, *Engineering atomic and molecular nanostructures at surfaces* Nature **437**, 671 (2005).
4. S. Facsko, T. Dekorsy, C. Koerdts, C. Trappe, H. Kurz, A. Vogt, and H. L. Hartnagel, *Formation of ordered nanoscale semiconductor dots by ion sputtering* Science **285**, 1551 (1999).
5. D. Flamm, F. Frost and D. Hirsch, *Evolution of surface topography of fused silica by ion beam sputtering* Appl. Surf. Sci. **179**, 95 (2001).
6. C. C. Umbach, R. L. Headrick, and K. C. Chang, *Spontaneous nanoscale corrugation of ion-eroded SiO₂: The role of ion-irradiation-enhanced viscous flow* Phys. Rev. Lett. **87**, 246104 (2001).
7. A. Toma, F. Buatier de Mongeot and U. Valbusa, *Ion beam erosion of amorphous materials: evolution of surface morphology* Nucl. Instrum. Methods Phys. Res. B **230**, 551 (2005).

8. J. Erlebacher, M. J. Aziz, E. Chason, M. B. Sinclair, and J. A. Floro, *Spontaneous pattern formation on ion bombarded Si(001)* Phys. Rev. Lett. **82**, 2330 (1999).
9. S. Habenicht, K. P. Lieb, J. Koch and A. D. Wieck, *Ripple propagation and velocity dispersion on ion-beam-eroded silicon surfaces* Phys. Rev. B **65**, 115327 (2002).
10. R. Gago, L. Vazquez, R. Cuerno, M. Varela, C. Ballesteros, and J. M. Albella, *Production of ordered silicon nanocrystals by low-energy ion sputtering* Appl. Phys. Lett. **78**, 3316 (2001).
11. B. Ziberi, F. Frost, T. Hoche, and B. Rauschenbach, *Ripple pattern formation on silicon surfaces by low-energy ion-beam erosion: Experiment and theory* Phys. Rev. B **72**, 235310 (2005).
12. A. D. Brown, J. Erlebacher, W. L. Chan, and E. Chason, *Transient topographies of ion patterned Si(111)* Phys. Rev. Lett. **95**, 056101 (2005).
13. C. L. Liu, J. M. Cohen, J. B. Adams, and A. F. Voter, *EAM study of surface self-diffusion of single adatoms of fcc metals Ni, Cu, Al, Ag, Au, Pd, and Pt* Surf. Sci. **253**, 334 (1991).
14. S. Rusponi, G. Costantini, C. Boragno, and U. Valbusa, *Ripple wave vector rotation in anisotropic crystal sputtering* Phys. Rev. Lett. **81**, 2735 (1998).
15. U. Valbusa, C Boragno and F. B. de Mongeot, *Nanostructuring surfaces by ion sputtering* J. Phys.: Condens. Matter **14**, 8153 (2002).
16. W. L. Chan, N. Pavenayotin, and E. Chason, *Kinetics of ion-induced ripple formation on Cu(001) surfaces* Phys. Rev. B **69**, 245413 (2004).

17. P. Karmakar and D. Ghose, *Ion beam sputtering induced ripple formation in thin metal films* Surf. Sci. Lett. **554**, L101 (2004).
18. H. Hansen, A. Redinger, S. Messlinger, G. Stoian, Y. Rosandi, H. M. Urbassek, U. Linke and T. Michely, *Mechanisms of pattern formation in grazing-incidence ion bombardment of Pt(111)* Phys. Rev. B **73**, 235414 (2006).
19. P. Sigmund, *Theory of Sputtering. I. Sputtering Yield of Amorphous and Polycrystalline Targets* Phys. Rev. **184**, 383 (1969)
20. M. A. Makeev, R. Cuerno and A.-L. Barabási, *Morphology of ion-sputtered surfaces* Nuc. Inst. and Methods. Phys. Res. B **197**, 185 (2002).
21. R. M. Bradley and J. M. E. Harper, *Theory of ripple topography induced by ion-bombardment* J. Vac. Sci. Technol. A **6**, 2390 (1988).
22. W.W. Mullins, J. Appl. Phys. **28**, 333 (1957); *Flattening of a Nearly Plane Solid Surface due to Capillarity* ibid **30**, 77 (1959).
23. C. Herring, in *Structure and Property of Solid Surfaces*, edited by R. Gomer and C. S. Smith (University of Chicago, Chicago, 1953).
24. A.-L. Barabási and H. E. Stanley, *Fractal Concepts in Surface Growth*, (Cambridge University Press, Cambridge, 1995).
25. A. Pimpinelli and J. Villain, *Physics of Crystal Growth* (Cambridge, 1998).
26. R. Cuerno and A.-L. Barabási, *Dynamic scaling of ion-sputtered surfaces* Phys. Rev. Lett. **74**, 4746 (1995).
27. I. Koponen, M. Hautala and O-P. Sievaenen, *Simulations of Ripple Formation on Ion-Bombarded Solid Surfaces* Phys. Rev. Lett. **78**, 2612 (1997).

28. K. Wittmaack, *Effect of surface roughening on secondary ion yields and erosion rates of silicon subject to oblique oxygen bombardment* J. Vac. Sci. Technol. A **8**, 2246 (1990).
29. J. Erlebacher, M. J. Aziz, E. Chason, M. B. Sinclair and J. A. Floro, *Nonclassical smoothening of nanoscale surface corrugations* J. Vac. Sci. Technol. A **18**, 115 (2000).
30. J. J. Vajo, R. E. Doty, and E-H. Cirlin, *Influence of O_2^+ energy, flux, and fluence on the formation and growth of sputtering-induced ripple topography on silicon* J. Vac. Sci. Technol. A **6**, 76 (1988).
31. E. A. Eklund, R. Bruinsma, J. Ruidnick, and R. S. Williams, *Submicron-scale surface roughening induced by ion bombardment* Phys. Rev. Lett. **67**, 1759 (1991).
32. H-N. Yang, G-C. Wang, and T-M. Lu, *Anomalous dynamic scaling on the ion-sputtered Si(111) surface* Phys. Rev. B **50**, 7635 (1994).
33. M. A. Makeev and A.-L. Barabási, *Ion-induced effective surface diffusion in ion sputtering* Appl. Phys. Lett. **71**, 2800 (1997).
34. S. Park, B. Kahng, H. Jeong, and A. L. Barabási, *Dynamics of ripple formation in sputter erosion: Nonlinear phenomena* Phys. Rev. Lett. **83**, 3486 (1999).
35. D. E. Wolf, *Kinetic roughening of vicinal surfaces* Phys. Rev. Lett. **67**, 1783 (1991).
36. M. Rost and J. Krug, *Anisotropic Kuramoto-Sivashinsky equation for surface growth and erosion* Phys. Rev. Lett. **75**, 3894 (1995).

37. S. Facsko, T. Dekorsy, and H. Kurz, *Energy dependence of quantum dot formation by ion sputtering* Phys. Rev. B **63**, 165329 (2001).
38. F. Frost, A. Schindler, and F. Bigl, *Roughness evolution of ion sputtered rotating InP surfaces: Pattern formation and scaling laws* Phys. Rev. Lett. **85**, 4116 (2000).
39. J. Muñoz-García, M. Castro, and R. Cuerno, *Nonlinear ripple dynamics on amorphous surfaces patterned by ion beam sputtering* Phys. Rev. Lett. **96** 086101 (2006).
40. M. Castro, R. Cuerno, L. Vázquez, and R. Gago, *Self-organized ordering of nanostructures produced by ion-beam sputtering* Phys. Rev. Lett. **94**, 016102 (2005).
41. O. Terzidis, P. Claudin, and J.-P. Bouchaud, *A model for ripple instabilities in granular media* Eur. Phys. J. B **5**, 245 (1998).
42. A. Valance and F. Rioual, *A nonlinear model for aeolian sand ripples* Eur. Phys. J. B **10**, 543 (1999).
43. Z. Csahók, C. Misbah, F. Rioual, and A. Valance, *Dynamics of aeolian sand ripples* Eur. Phys. J. E **3**, 71 (2000).
44. F. Frost, B. Ziberi, T. Hoche, and B. Rauschenbach, *The shape and ordering of self-organized nanostructures by ion sputtering* Nucl. Instrum. Methods Phys. Res. B **216**, 9 (2004).
45. G. Carter, *Proposals for producing novel periodic structures by ion bombardment sputtering* Vacuum **77**, 97 (2004); *Surface ripple amplification and attenuation by sputtering with diametrically opposed ion fluxes* *ibid* **79**, 106 (2005).

46. S. Vogel and S. J. Linz, *Surface structuring by multiple ion beams* Phys. Rev. B **75**, 085425 (2007).
47. B. B. Mandelbrot, *The Fractal Geometry of Nature*, (Freeman, New York, 1982).
48. J. Krim and G. Palasantzas, *Experimental Observations of Self-Affine Scaling and Kinetic Roughening at Sub- Micron Lengthscales*, International Journal of Modern Physica B **9**, 599-632 (1995).
49. P. Sigmund, *A mechanism of surface micro-roughening by ion bombardment* J. Mat. Sci. **8**, 1545 (1973).
50. S.-P. Kim, J.-S. Kim, and K.-R. Lee, (unpublished).
51. N. Toyoda, T. Mashita, and I. Yamada, *Surface smoothing mechanism of gas cluster ion beams* Nucl. Instrum. Methods Phys. Res. B **161-163**, 980 (2000).
52. F. Family and T. Vicsek, *Scaling of the active zone in the Eden process on percolation networks and the ballistic deposition model* J. Phys. A **18**, L75 (1985).
53. I. Elkinani and J. Villain, *Le paradoxe de zenon d'eele* Solid State Commun. **87**, 105 (1993).
54. B. Ziberi, F. Frost, M. Tartz, H. Neumann, and B. Rauschenbach *Ripple rotation, pattern transitions, and long range ordered dots on silicon by ion beam erosion* Appl. Phys. Lett. **92** (6) (2008).
55. H. Jeong, B. Kahng, D. Kim, *Dynamics of a Toom interface in three dimensions* Phys. Rev. Lett. **71**, 747 (1993).
56. Z. X. Liu and P. F. A. Alkemade, *Flux dependence of oxygen-beam-induced ripple growth on silicon* Appl. Phys. Lett. **79**, 4334 (2001).

57. G. Costantini, E. B. de Mongeot, C. Boragno, and U. Valbusa, *Is Ion Sputtering Always a “Negative Homoepitaxial Deposition”?* Phys. Rev. Lett. **86**, 838 (2001).
58. E. Carrasco, O. Rodríguez de la Fuente, M. A. González, and J. M. Rojo, *Characterising and controlling surface defects* Eur. Phys. J. B **40**, 421 (2004).
59. H. Zhou, Y. P. Wang, L. Zhou, R. L. Headrick, A. S. Ozcan, Y. Y. Wang, G. Ozaydin, K. F. Ludwig, and D. P. Siddons, *Wavelength tunability of ion-bombardment-induced ripples on sapphire* Phys. Rev. B **75**, 155416 (2007).
60. P. Mishra and D. Ghose, *The energy dependence of sputtering induced ripple topography in Al film* Nucl. Instrum. Methods Phys. Res. B **1635**, 266 (2008).
61. M. Joe, C. Choi, B. Kahng, and J.-S. Kim, *Nanopatterning by dual-ion-beam sputtering* Appl. Phys. Lett. **91**, 233115 (2007).
62. The referred f is the nominal one estimated from the integrated flux over the entire irradiated area that extends further beyond the sample area due to both the grazing incidence and the defocused operation of the ion gun for the purpose of uniform irradiation of the sample.
63. U. Valbusa, C. Boragno, and F. B. de Mongeot, *Nanostructuring by ion beam* Mater. Sci. Eng. C **23**, 201 (2003).
64. B. Ziberi, F. Frost, B. Rauschenbach, and T. Hoche, *Highly ordered self-organized dot patterns on Si surfaces by low-energy ion-beam erosion* Appl. Phys. Lett. **87** 033113 (2005).
65. F. Frost, R. Fechner, D. Flamm, B. Ziberi, W. Frank, and A. Schindler, *Ion beam assisted smoothing of optical surfaces* Appl. Phys. A: Mater. Sci. Process. **A78**, 651 (2004).

66. T. C. Kim, M. H. Jo, Y. Kim, D. Y. Noh, B. Kahng, and J.-S. Kim, *Morphological evolution of ion-sputtered Pd(001): Temperature effects* Phys. Rev. B **73**, 125425 (2006).
67. O. Malis, J. D. Brock, R. L. Headrick, M.-S. Yi, and J. M. Pomeroy, *Ion-induced pattern formation on Co surfaces: An x-ray scattering and kinetic Monte Carlo study* Phys. Rev. B **66**, 035408 (2002).
68. M. Joe, C. Choi, B. Kahng, C. Y. Kwak, and J.-S. Kim, *Patterning by ion-beam sputtering* J. Korean Phys. Soc. **52**, S181 (2008).
69. M. Kardar, G. Parisi, and Y.-C. Zhang, *Dynamic scaling of growing interfaces* Phys. Rev. Lett. **56**, 889 (1986).
70. <http://www.srim.org>; J. F. Ziegler, J. P. Biersack, and U. Littmark, *The Stopping and Range of Ions in Solids* (Pergamon, New York, 1985).
71. K. Wittmaack, *Reliability of a popular simulation code for predicting sputtering yields of solids and ranges of low-energy ions* J. App. Phys., **96**, 2632 (2004).
72. H. H. Chen, O. A. Urquidez, S. Ichim, L. H. Rodriguez, M. P. Brenner, and M. J. Aziz, *Shocks in ion sputtering sharpen steep surface features* Science **310**, 294 (2005).
73. T. C. Kim, C. M. Ghim, H. J. Kim, D. H. Kim, D. Y. Noh, N. D. Kim, J. W. Chung, J. S. Yang, Y. J. Chang, T. W. Noh, B. Kahng and J.-S. Kim, *Kinetic roughening of ion-sputtered Pd(001) surface: Beyond the Kuramoto-Sivashinsky model* Phys. Rev. Lett., **92**, 246104 (2004).
74. L. Vitos, A. V. Ruban, H. L. Skriver, and J. Koll, *The surface energy of metals* Surf. Sci. **411**, 186 (1998).

75. W. L. Chan and E. Chason, *Making waves: Kinetic processes controlling surface evolution during low energy ion sputtering* J. Appl. Phys. **101**, 121301 (2007).
76. S. K. Mohanta and R. K. Soni, *Structural modifications in InP nanostructures prepared by Ar⁺-ion irradiation* J. Appl. Phys. **102**, 074313 (2007).
77. M. Lu, X. J. Yang, S. S. Perry, and J. W. Rabalais, *Self-organized nanodot formation on MgO(100) by ion bombardment at high temperatures* Appl. Phys. Lett. **80**, 2096 (2002).
78. G. Carter, M. J. Nobes, and J. L. Whitton, *Sputtering induced topography development on f.c.c. metals* Appl. Phys. A **38**, 77 (1985).
79. J. T. Drotar, Y. P. Zhao, T. M. Lu, and G. C. Wang, *Numerical analysis of the noisy Kuramoto-Sivashinsky equation in 2+1 dimensions* Phys. Rev. E **59**, 177 (1999).
80. S. Habenicht, W. Bolse, K. P. Lieb, K. Reimann, and U. Geyer, *Nanometer ripple formation and self-affine roughening of ion-beam-eroded graphite surfaces* Phys. Rev. B **60**, R2200 (1999).
81. Q. Wei, J. Lian, S. Zhu, W. Li, K. Sun, and L. Wang, *Ordered nanocrystals on argon ion sputtered polymer film* Chem. Phys. Lett. **452**, 124 (2008).
82. J.-H. Kim, M. Joe, S.-P. Kim, N.-B. Ha, B. Kahng, K.-R. Lee, and J.-S. Kim, (unpublished).
83. J.-H. Kim, N.-B. Ha, and J.-S. Kim, (unpublished).
84. A. Cuenat, H. B. George, K. C. Chang, J. M. Blakely, and M. J. Aziz, *Lateral templating for guided self-organization of sputter morphologies* Adv. Mater. **17**, 2845 (2005).

85. B. D. Yu and M. Scheffler, *Physical origin of exchange diffusion on fcc(100) metal surfaces* Phys. Rev. B **56**, R15569 (1997).
86. T. Aste and U. Valbusa, *Surface instabilities in granular matter and ion-sputtered surfaces* Physica A (Amsterdam) **332**, 548 (2004).
87. R. Gago, L. Vázquez, R. Cuerno, C. Ballesteros and J. M. Albella, *Nanopatterning of silicon surfaces by low-energy ion-beam sputtering: dependence on the angle of ion incidence* Nanotechnology **13**, 304 (2002).
88. W. H. Press, B. P. Flannery, S. A. Teukolsky and W. T. Vetterling, *Numerical Recipes* (Cambridge Univ. Press, Cambridge, 1986).
89. B. Kahng, H. Jeong, and A.-L. Barabási, *Quantum dot and hole formation in sputter erosion* Appl. Phys. Lett. **78**, 805 (2001).
90. B. Kahng, H. Jeong, and A.-L. Barabási, *Nanoscale pattern formation in sputter eroded surface* J. Korean Phys. Soc. **39**, 421 (2001).
91. S. Habenicht, W. Bolse, H. Feldermann, U. Geyer, H. Hofsass, K. P. Lieb, and F. Roccaforte, *Ripple topography of ion-beam-eroded graphite: A key to ion-beam-induced damage tracks* Europhys. Lett. **50** 209 (2000).
92. J. Kim, B. Kahng and A.-L. Barabási, *Nanoscale wire formation on sputter-eroded surfaces* Appl. Phys. Lett. **81**, 3654 (2002).
93. W. B. Fan, L. J. Qi, H. T. Sun, Y. Y. Zhao, and M. Lu, *Modification of Si nanostructures on Si(100) by post ion milling* Nanotechnology **17**, 1878 (2006).
94. W. B. Fan, L. Ling, L. J. Qi, W. Q. Li, H. T. Sun, C. X. Gu, Y. Y. Zhao, and M. Lu, *Modification of Si nanostructures on Si(100) by post ion milling* J. Phys. Condens. Matter **18**, 3367 (2006).

95. S. Facsko, T. Bobek, A. Stahl, H. Kurz and T. Dekorsy, *Dissipative continuum model for self-organized pattern formation during ion-beam erosion* Phys. Rev. B **69**, 153412 (2004).
96. S. M. Foiles, M. I. Baskes and M. S. Daw, *Embedded-atom-method functions for the fcc metals Cu, Ag, Au, Ni, Pd, Pt, and their alloys* Phys. Rev. B **33**, 7983 (1986).
97. J. F. Ziegler, J. P. Biersack, U. Littmark, *The Stopping and Range of Ions in Solids*, (Pergamon, New York 1985).
98. M. A. Karolewski, *Kalypso: a software package for molecular dynamics simulation of atomic collisions at surfaces* Nucl. Instrum. Methods Phys. Res. B **230**, 402 (2005).
99. J. D. Kress, D. E. Hanson, A. F. Voter, C. L. Liu, X.-Y. Liu, and D. G. Coronell, *Molecular dynamics simulation of Cu and Ar ion sputtering of Cu (111) surfaces* J. Vac. Sci. Technol. A **17**, 2819 (1999).
100. A. Rettori, and J. Villain, *Flattening of grooves on a crystal surface: a method of investigation of surface roughness* J. Phys. (Paris) **49**, 257 (1988).
101. M. Ozdemir, and A. Zangwill, *Morphological equilibration of a corrugated crystalline surface* Phys. Rev. B **42**, 5013 (1990).
102. I. M. Lifshitz and V. V. Slyozov, *The kinetics of precipitation from supersaturated solid solutions* J. Phys. Chem. Solids **19**, 35 (1961).
103. I. M. Lifshitz, Sov. Phys. JETP **12**, 939 (1962); S. M. Allen and J. W. Cahn, *A microscopic theory for antiphase boundary motion and its application to antiphase domain coarsening* Acta Metall. **27**, 1085 (1979).

- 104. K. Kern, I. K. Robinson, and E. Vlieg, *Healing kinetics of a sputter-roughened surface* Surf. Sci. **261**, 118 (1992).
- 105. J. Erlebacher, M. J. Aziz, E. Chason, M. B. Sinclair, and J. A. Floro, *Non-classical smoothening of nanoscale surface corrugations* Phys. Rev. Lett. **84**, 5800 (2000).
- 106. G. Carter, V. Vishnyakov, and M. J. Nobes, *Ripple topography development on ion bombarded Si* Nucl. Instrum. Methods Phys. Res. B **115**, 440 (1996).

초록

이 논문에서는 이온 다발 때려내기를 이용한 나노 패턴 형성에 관한 실험 및 이론 연구를 수행하였다. 실험적 접근으로써, Au(001) 표면에 비스듬한 이온 다발을 입사하여 때려내기를 하였다.

첫째로, 홀 이온 다발 때려내기를 이용하여 표면에 리플 구조를 만들었고, 이러한 패턴은 여러가지 때려내기 조건, 예를 들면 이온 다발의 플렉스, 플루언스, 에너지, 결정기관의 결정축과 이온 다발 쏘비축이 평면에서 이루는 각도등에 의존한다. 이러한 때려내기 맺음변수등을 조절함으로써 다양한 형태의 리플 구조물을 얻을 수 있다. 잘 정의된 리플 구조물을 얻기 위해, 표면 퍼짐에의한 질서 저하를 극복하는 충분한 이온 플렉스와 아울러 결정축과 나란한 입사조건이 요구된다. 이온-침식 동역학과 표면 퍼짐 동역학 사이의 경쟁 때문에 금 단결정기관위에 다소 풍부한 리플 상들이형성된다.

둘째로, 겹 이온 다발 때려내기를 이용하여 홀 이온 다발 때려내기보다 다양한 패턴 형성 가능성을 연구하였다. 겹 이온 다발 때려내기 방법을 사용하여 잘 배열된 이차원 나노패턴을 만들었다. 또한 여러 새로운 특징들이 발견되었다. 예를 들면, 리플의 사면의 비대칭성과 플렉스로 인해 생기는 표면 불안정성등이다.

셋째로, 교차 이온 다발 때려내기를 이용하여 미리 만들어진 리플의 변형을 관찰하였다. 처음 리플의 변형과 새로 형성되는 리플을 통해서 때려내기 과정에 관여하는 동역학을 알아낼 수 있다. 브래들리-하퍼의 열적 평탄화와 달리, 재증착과 기울기에 의존하는 침식등의 비열적 과정이 평탄화를 일으킬 수 있다. 처음 리플의 가로 크기 성장은 열적 성장에 비해 꽤 큰 성장지수를

가지고 성장한다. 따라서, 새로 생기는 리플에 의한 효과적인 질량 수송 메커니즘을 제안 하였다. 또한 이온 침식이 강한 영역에서는 초기에는 먹함수 거동을 보이다 나중에는 로그 함수 거동을 보이는 교차를 나타내는 축척 거동을 보인다. 로그 거동은 때려낸 원자들의 표면 재증착에 기인할 수 있다.

이론적 접근으로써, 겹 이온 다발 때려내기를 고려한 변형된 쿠라모토-시바신스키 방정식을 세웠다. 그러나 근본적으로 홀 이온 다발 때려내기를 설명하는 쿠라모토-시바신스키 방정식과 같았고, 표면에 패턴 형성을 제공하는 표면 불안정성 및 여러 특징들을 설명하지 못하였다.

때려낸 원자들의 재증착 가설을 확인하기 위해, 교차 이온 다발 때려내기를 모사하는 큰 규모의 분자동역학시뮬을 수행하였다. 비록 짧은 시뮬 시간이었음에도 불구하고, 리플의 골짜기와 그늘진 사면에 재증착이 일어났고 패턴 형성에 중요한 역할을 한다.

요약하자면, 겹 이온 다발을 사용하는 새로운 이온 때려내기 실험을 단결정 기관 위에 수행하여, 다양한 패턴을 얻었다. 또한 이러한 실험 결과를 설명할 수 있는 이론을 찾고자 하였다. 이러한 패턴형성은 종래의 이론이 설명할 수 있는 범위를 넘어서는 것이며, 계속해서 이론적인 연구가 진행중이다.

주요어: 패터닝, 패턴형성, 나노패터닝, 때려내기, 이온 다발 때려내기, 나노 구조물, 쿠라모토-시바신스키 방정식, 리플, 브래들리-하퍼 이론, 일차 불안정성 분석

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감사의 글

먼저, 저와 항상 함께 하여주시고 인생을 선하게 인도하시는 하나님께 모든 감사와 영광을 돌려드립니다. 석·박사 과정 동안 여러모로 부족한 저를 지도 해주시고, 믿고 지켜봐주시고, 지원해 주신 강병남 교수님께 진심으로 감사합니다. 선생님의 가르침을 발판으로 앞으로 훌륭한 과학자가 되도록 노력하겠습니다. 숙명여대 김재성 교수님께 진심으로 감사합니다. 선생님과 함께 하며 연구를 어떻게 하는지 배웠습니다. 또한 하나의 연구 결과가 노력과 인내로 말미암는다는 값진 교훈을 몸소 얻게 도와주셨습니다. 김두철 교수님에게 진심으로 감사합니다. “자네 논문은 자네가 할아버지가 되어도 남아있을 역사의 기록이네. 의미 있다고 생각하는 것을 잘 기록하게.” 격려 해주셔서 감사합니다. 선생님의 말씀을 잊지 않고 과학자의 길을 가겠습니다. 심사를 맡아 주신 존경하는 홍종배 교수님, 최석봉 교수님, 송실대 김진민 교수님들에게 진심으로 감사합니다. KIST 이광렬 박사님께 진심으로 감사합니다. 이 논문이 나오기까지 많은 배려를 해 주시고, 베스트의 연구 환경 가운데서 마음껏! 연구할 수 있는 기회를 주셔서 진심으로 감사합니다. 스페인의 Rodolfo Cuerno 교수님께 진심으로 감사합니다. 짧은 기간이었지만 함께하면서 늘 저의 얘기를 겸손히 들어주시고 지도해 주셨습니다. 크고 작은 많은 도움을 주신 KIST 문명운 박사님, 김상필 박사님께 진심으로 감사합니다. 늘 저를 힘써 도와준 숙명여대 김지현에게 진심으로 감사합니다. 통계물리그룹의 박지용 박사님, 김진섭, 이재성, 윤창근, 노정호, 김강훈, 최철호에게 진심으로 감사합니다. 똑똑하고, 유머러스한 동료들과 함께 하면서 즐겁게 물리를 공부할 수 있었습니다. 끝으로, 물심양면으로 저를 섬겨주시고 기도해주신 가족들께 진심으로 감사합니다.